



Alternative Data Mining and Its Application (in Finance)

另类数据挖掘及其(在金融中的)应用

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上海科技大学

- **What:** 什么是另类数据？
- **Why:** 我们为什么要利用另类数据？
- **How:** 我们如何使用另类数据？

- **Beyond the typical company filings, earnings calls, or fundamental datasets ...** Investors are looking for new signals which may give them an edge over their competitors. – Forbes

超越传统的公司财报、业绩会议、基本面等金融数据，投资者正在寻找能够帮助他们占得先机的新数据。（福布斯）

200年前的例子

- 19世纪，伦敦霍乱造成四万多人死亡，学界普遍认为瘟疫来源于【瘴气】
- 年轻医生John Snow有不同的观点

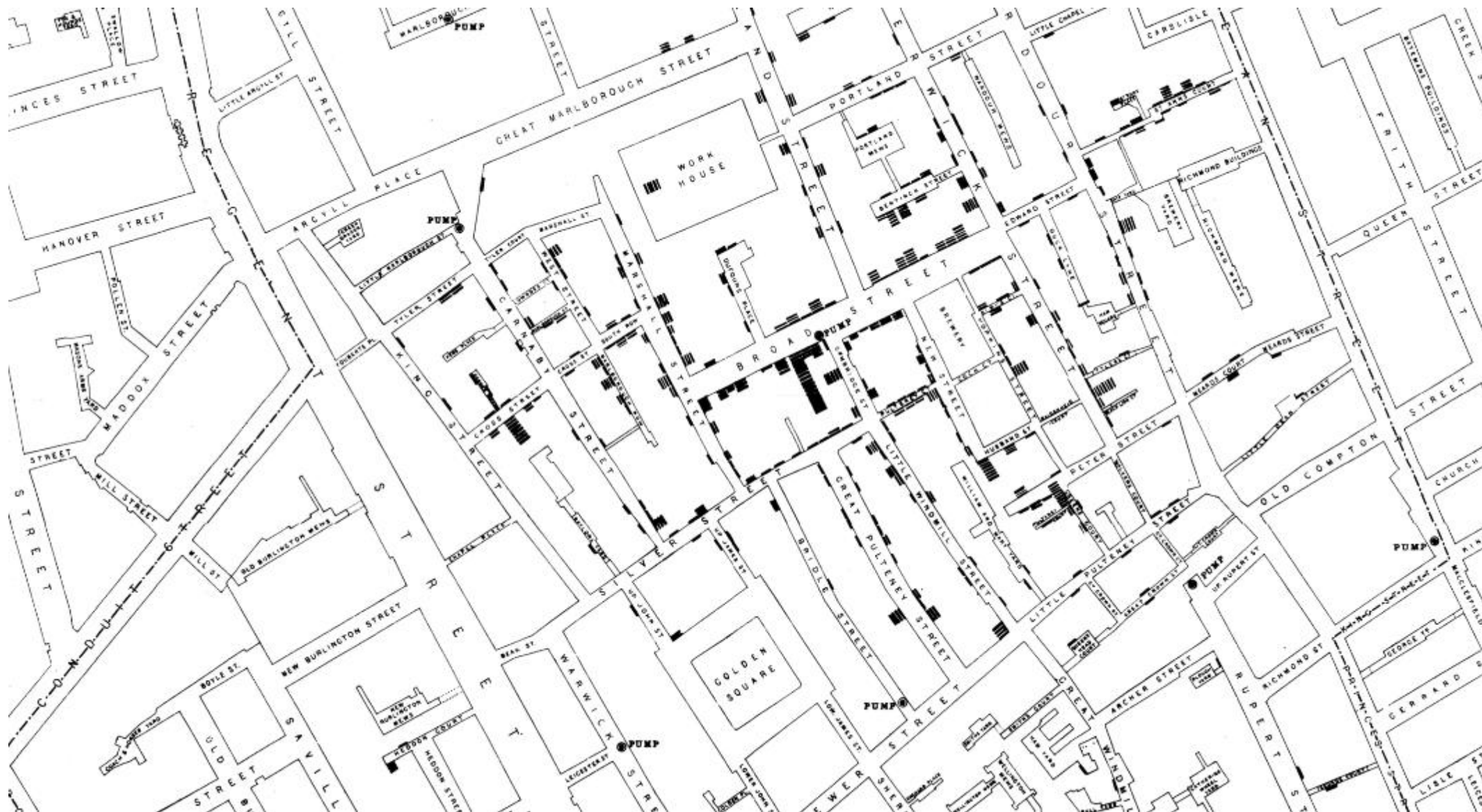


A COURT FOR KING CHOLERA.



John Snow

- Snow绘制疫情地图

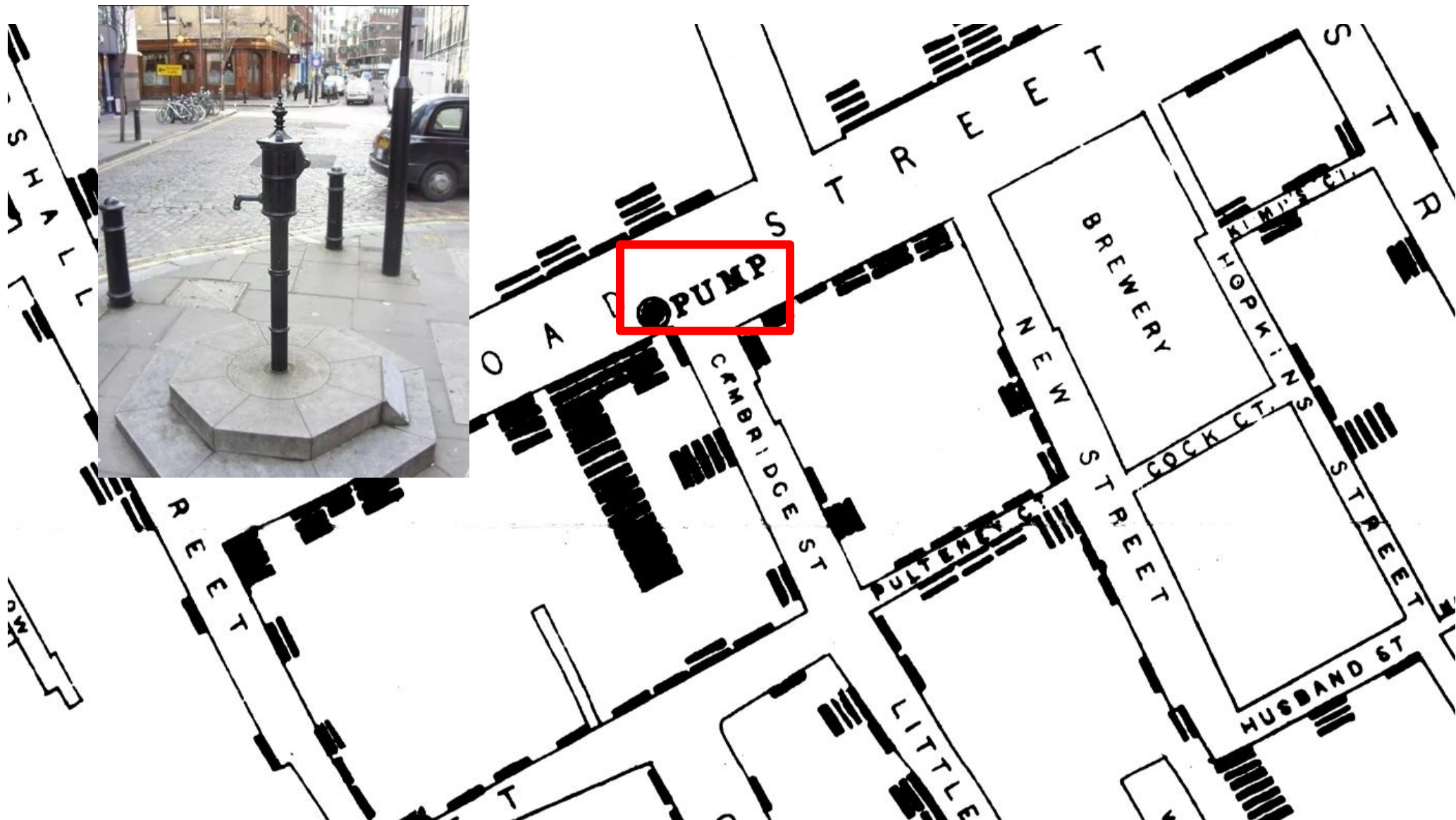


- Snow绘制疫情地图





拆除被污染水泵把手，疫情得到遏制

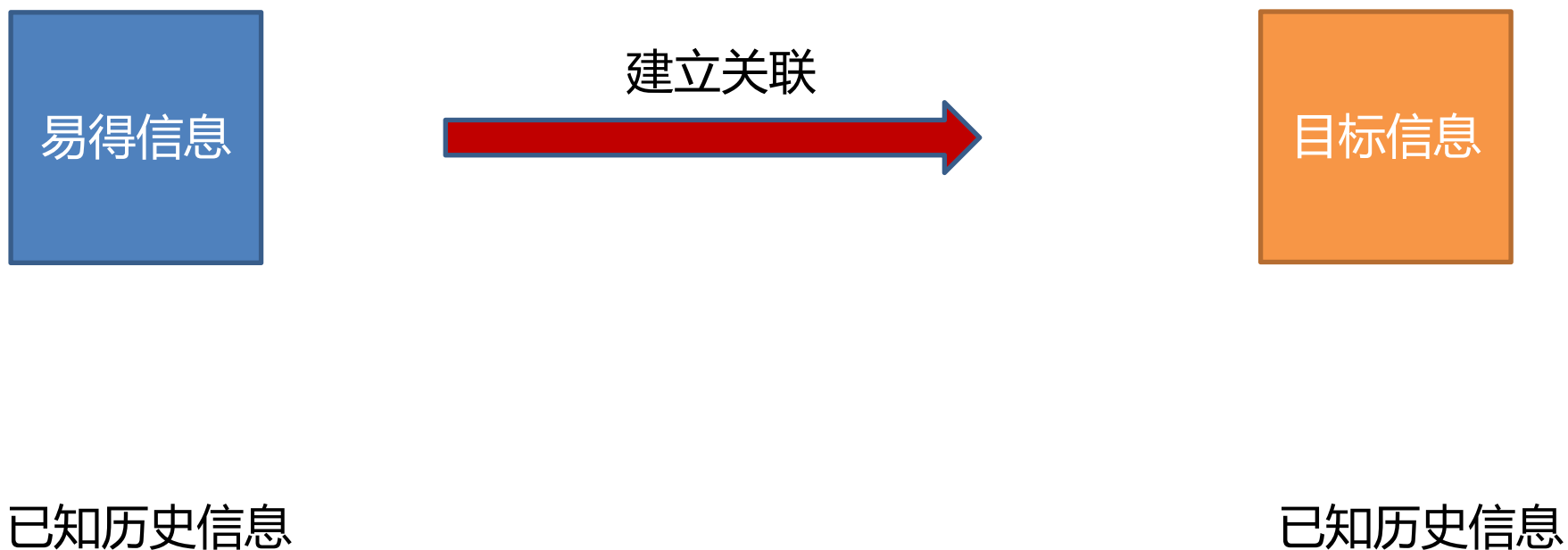


建设现代城市下水道系统 (1859)



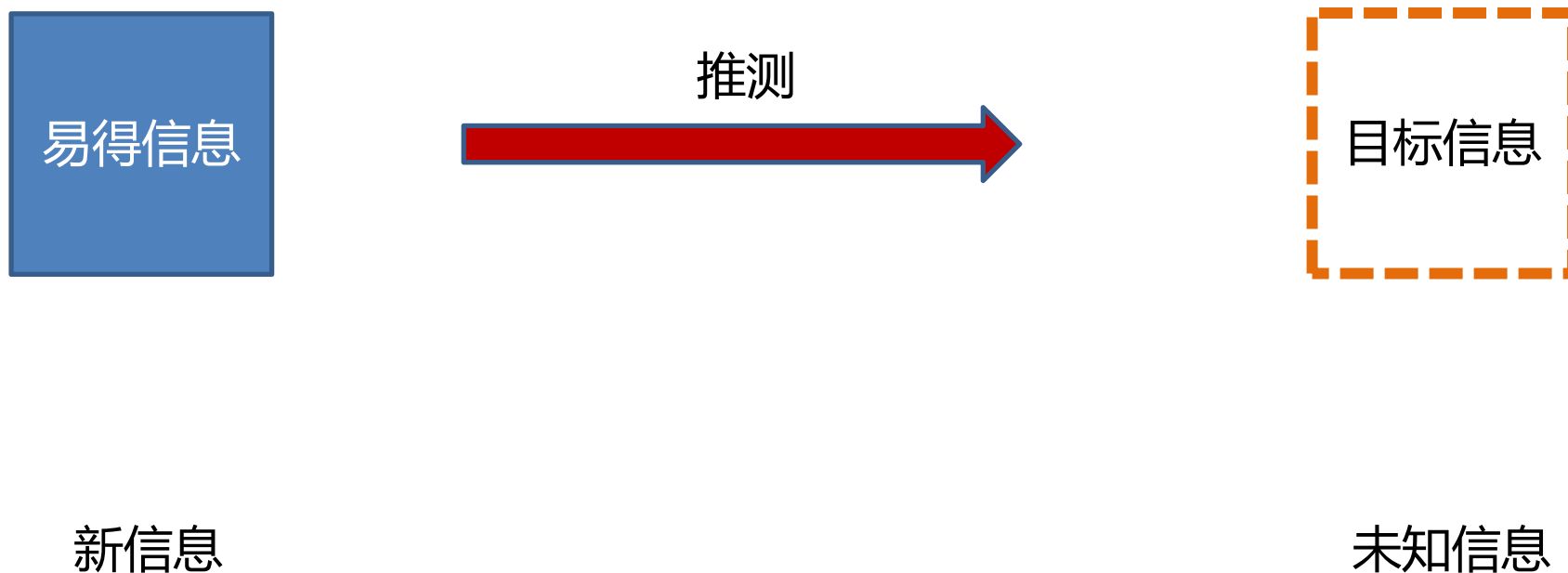
数据挖掘的思路

- 寻找已知信息间的关联



数据挖掘的思路

- 寻找已知信息间的关联
- 推测未知信息



IBM Project Debater

- Nature封面文章 (18 March, 2021)
- 人工智能系统与人类进行辩论赛
- 项目始于2012年

Slonim, Noam, et al. "An autonomous debating system." *Nature* 591.7850 (2021): 379-384.



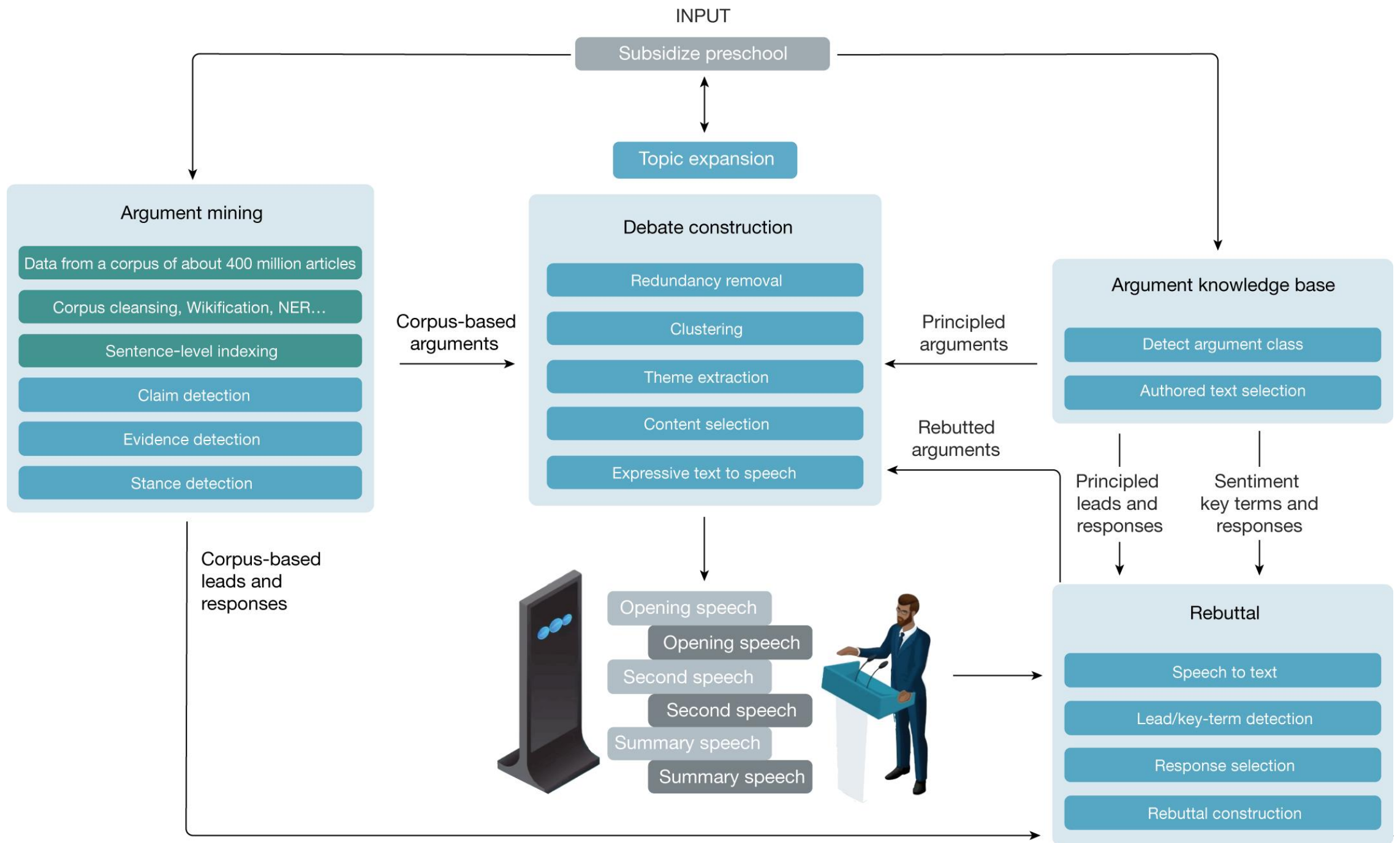
Opening
Rebuttal
Summary

03:50:04

Project Debater

Noa Ovadia





另类数据挖掘及其在金融中的应用

- **另类数据及其商业化**
 - Web/App/Social Media
 - Satellite
 - Location/Foot Traffic
- **另类数据与现实世界**
 - 挖掘时间地理数据对生态现象建模
- **另类数据与人类行为**
 - 手机数据与用户特征
 - 社交媒体与电子商务用户行为
- **另类数据与经济金融研究**
 - 挖掘卫星图片预测经济指标
 - 突发风险事件监测与市场建模

Web/App/Social Media

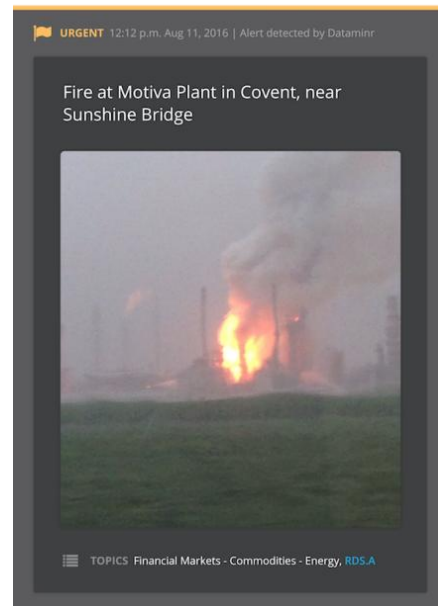
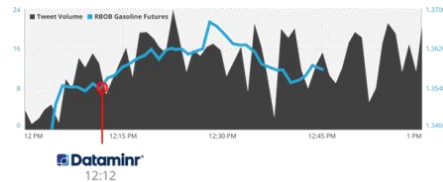
- **从网络、APP、社交媒体数据中发现信号。**
 - [AppAnnie](#), 收集App store中的表现与app中的数据, 进行市场分析。
 - [7ParkData](#), 收集App安装数据, 消费数据, 网站流量数据, 评估公司表现。
 - [Dataminr](#), 与Twitter合作, 实时发现重要事件, 转化为金融、公司舆情、市政响应的信号。

Dataminr

- 在主流媒体之前发现炼油厂大火、食品安全问题。

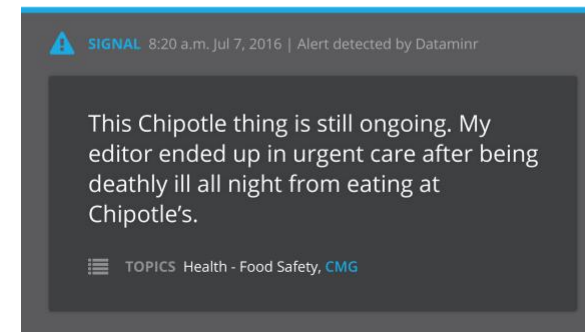
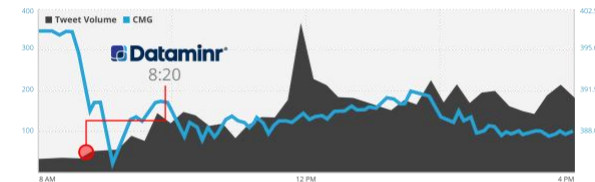
Motiva Refinery Fire

On August 11, Dataminr surfaced an image posted to Twitter of a blaze at a crude oil refinery in Covent, Louisiana. Dataminr notified clients of the fire in advance of both major news and a 5% spike in U.S. Distillate futures. The plant, owned by Motiva Enterprises in a 50-50 joint venture between Shell Oil Company and Saudi Refining (controlled by Saudi Aramco), was taken partially off-line following the fire, significantly impacting production at this 225,000-barrel-per-day refinery.



Chipotle Food Safety Rumors

On July 7, Dataminr's algorithm detected a Tweet indicating that Chipotle's long string of food safety concerns was far from resolved, notifying users over one hour before Chipotle's (CMG) stock price tumbled nearly 3.5% at market open. The rumors reignited concerns about the chain's food safety practices following multiple E. coli outbreaks earlier this year.



Satellite

- 卫星图像中的识别，如沃尔玛门前车辆数，农产品的色泽，预测商品和宏观经济。
 - [Rezatec](#), 遥感监控环境，用于农业、水资源、林业、能源领域。
 - [Windward](#), 监控船只，了解海运贸易的趋势。
 - [Orbital Insight](#), 识别卫星及无人机照片中的车辆数、建筑阴影高度（建筑高度）、雾霾等。

Orbital Insight

- 计算机视觉方法提取出的南京某卫星图片中的阴影



Orbital Insight

- 中国经济指标，追踪30+个城市。
 - **City Economic Development**, car density, truck density, building construction, electricity consumption (nighttime illumination), road building
 - **Import and Export**, port activity (ship count and sizes) , rail throughput, aircraft count, commodity stockpiles at ports
 - **Industrial and Rural Output**, agricultural yields, crop health, pit mining activity, factory activity (trucking activity)
 - **Energy**, oil storage, development of coal plants, solar cell farms, and wind farms

Location/Foot Traffic

- 从地点(check-in, 手机GPS)和视频数据中, 收集位置的人流信息。
 - [Foursquare](#), 社交网站, 允许用户在地点check-in并点评。曾预测Chipotle业绩不及预期。
 - [Placemeter](#), 采集视频, 分析视频中的车辆和行人数量, 用以评估地点的商业价值, 优化经营模式等。
 - [Placed](#), 收集用户的手机GPS数据, 评估地点。

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Mining 150 million social media photos, to model the real world

Zhang, H. et al. Mining photo-sharing websites to study ecological phenomena. In Proceedings of the 21st international conference on World Wide Web, ACM, pp. 749-758.

Media coverage:

Snow snaps give you a better weather picture, in [New Scientist](#),
in [ACM Communication Website](#)

- A Flickr photo



Scott Scheetz

Taken on September 22, 2012



Bloomington, Indiana, United States

Indiana Lotus Festival

Street Photography

Kirkwood Ave

Sample Gates

Indiana University

Wedding photographer

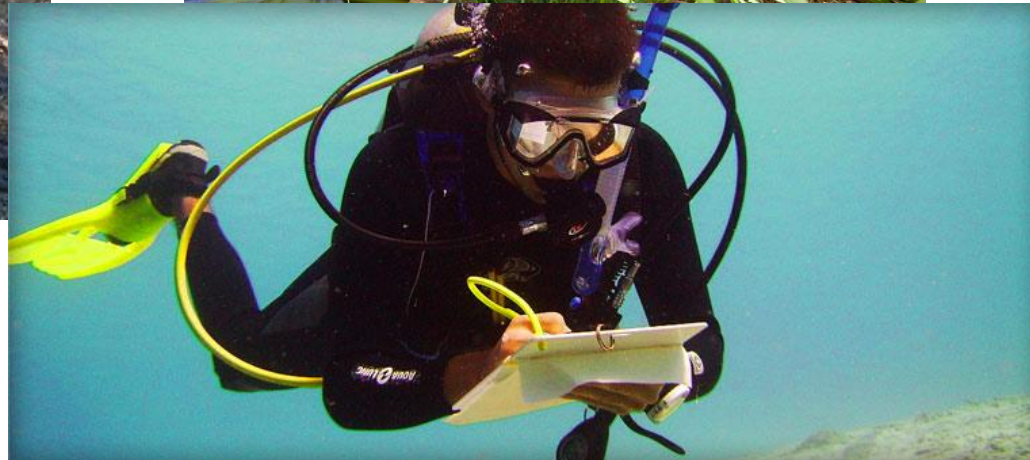
when

where

Who and what

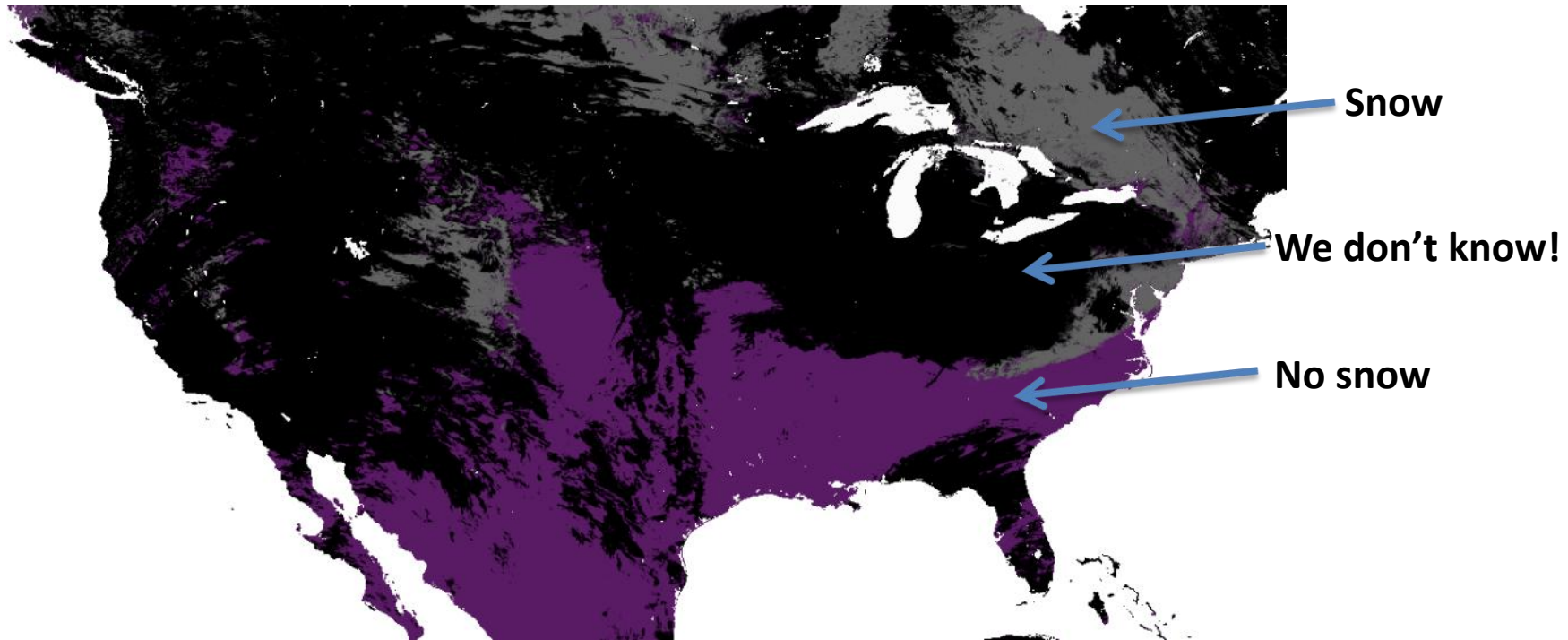
Need for data in biology and ecology

- Species distribution, animal migration
 - Field observation data



Need for data in biology and ecology

- Study of natural phenomena
 - Satellite data

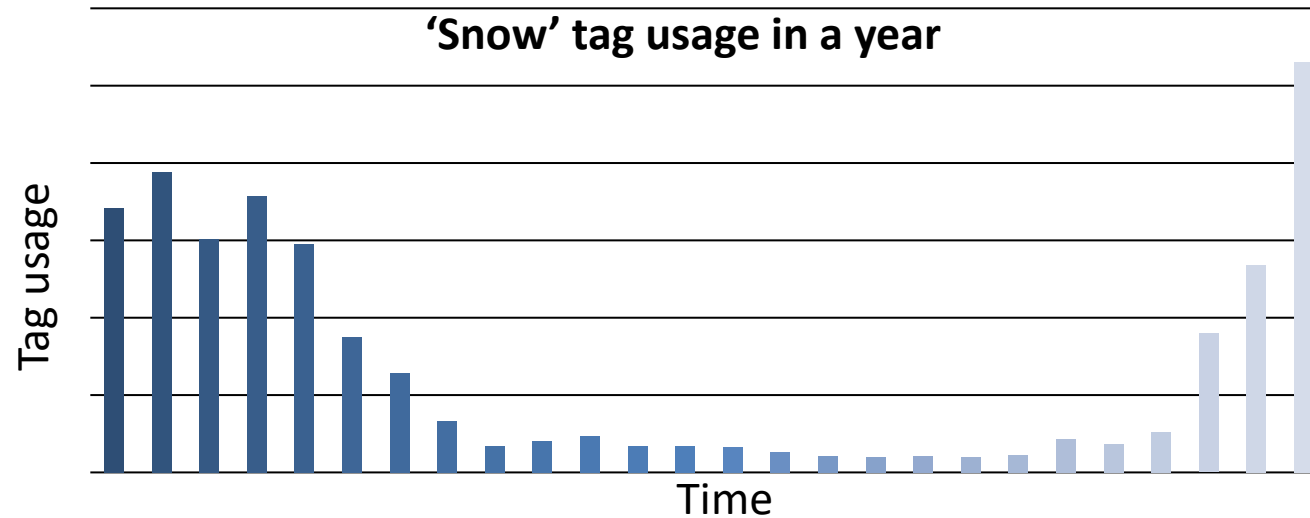
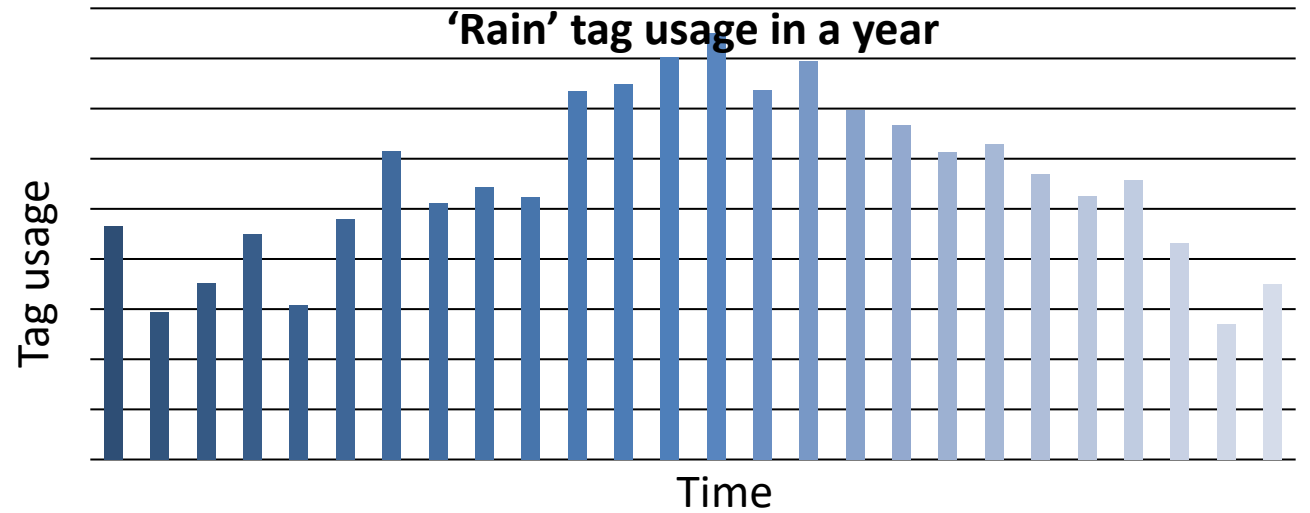


Can web data help?

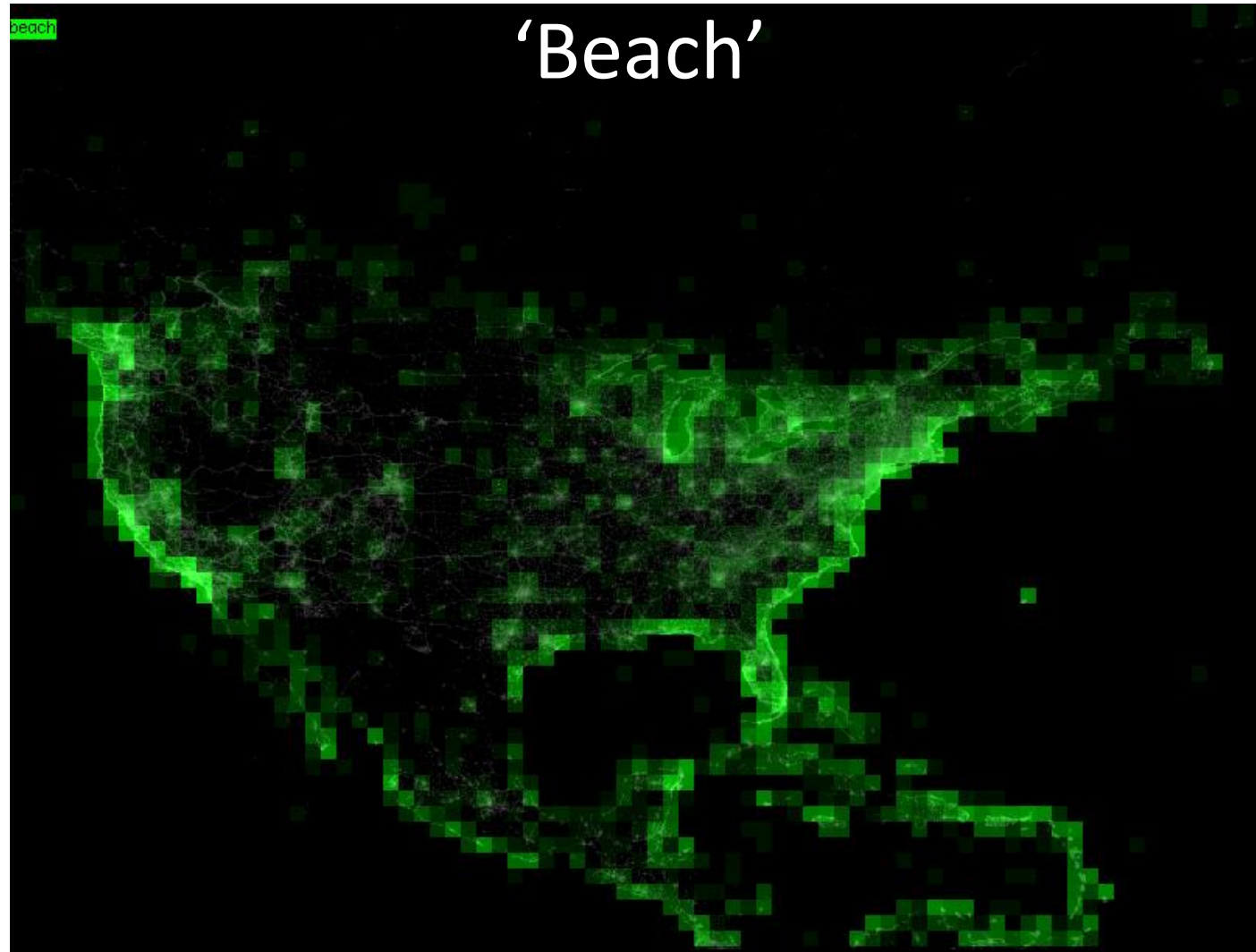
2018.5.20



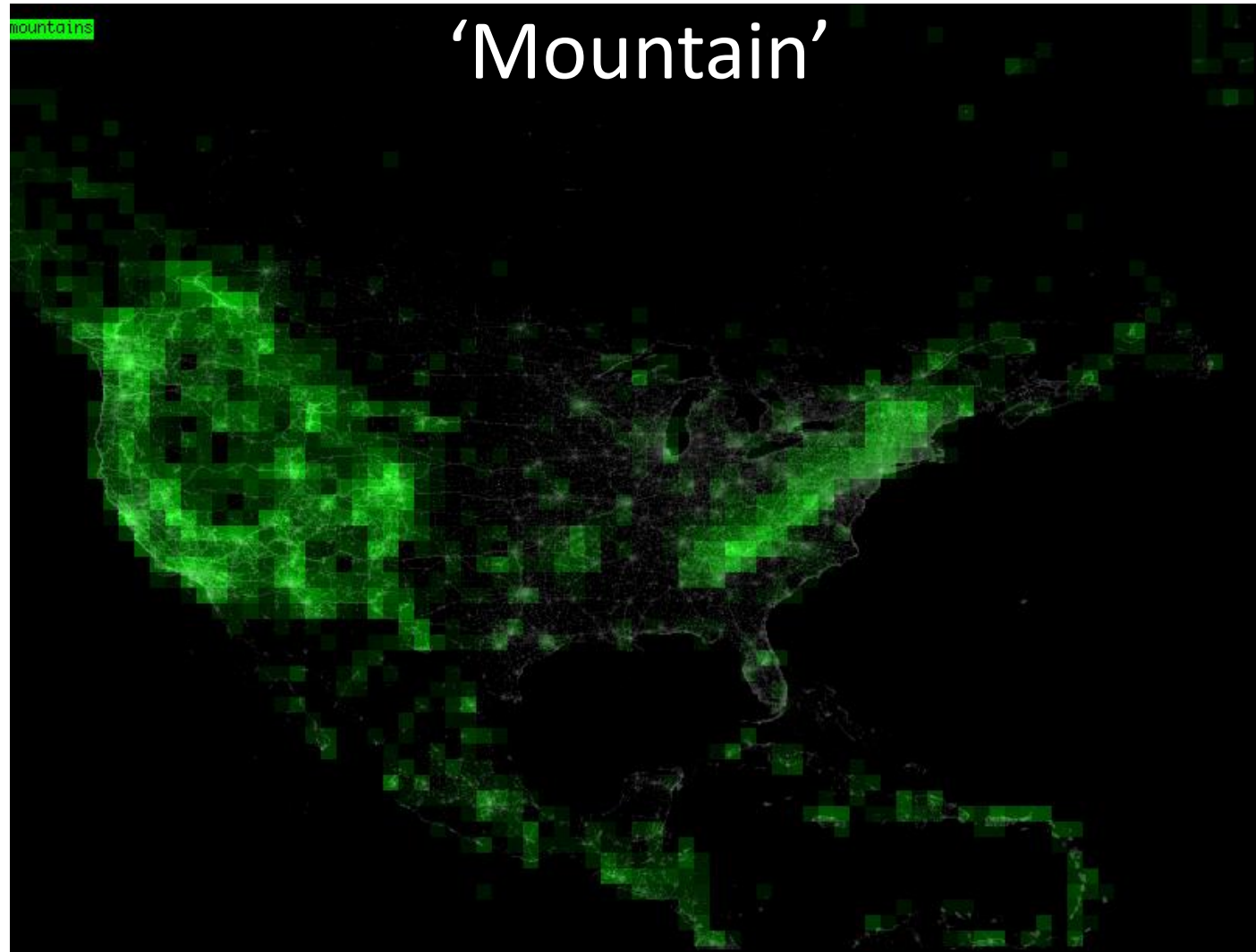
Data's temporal property



Data's geo property

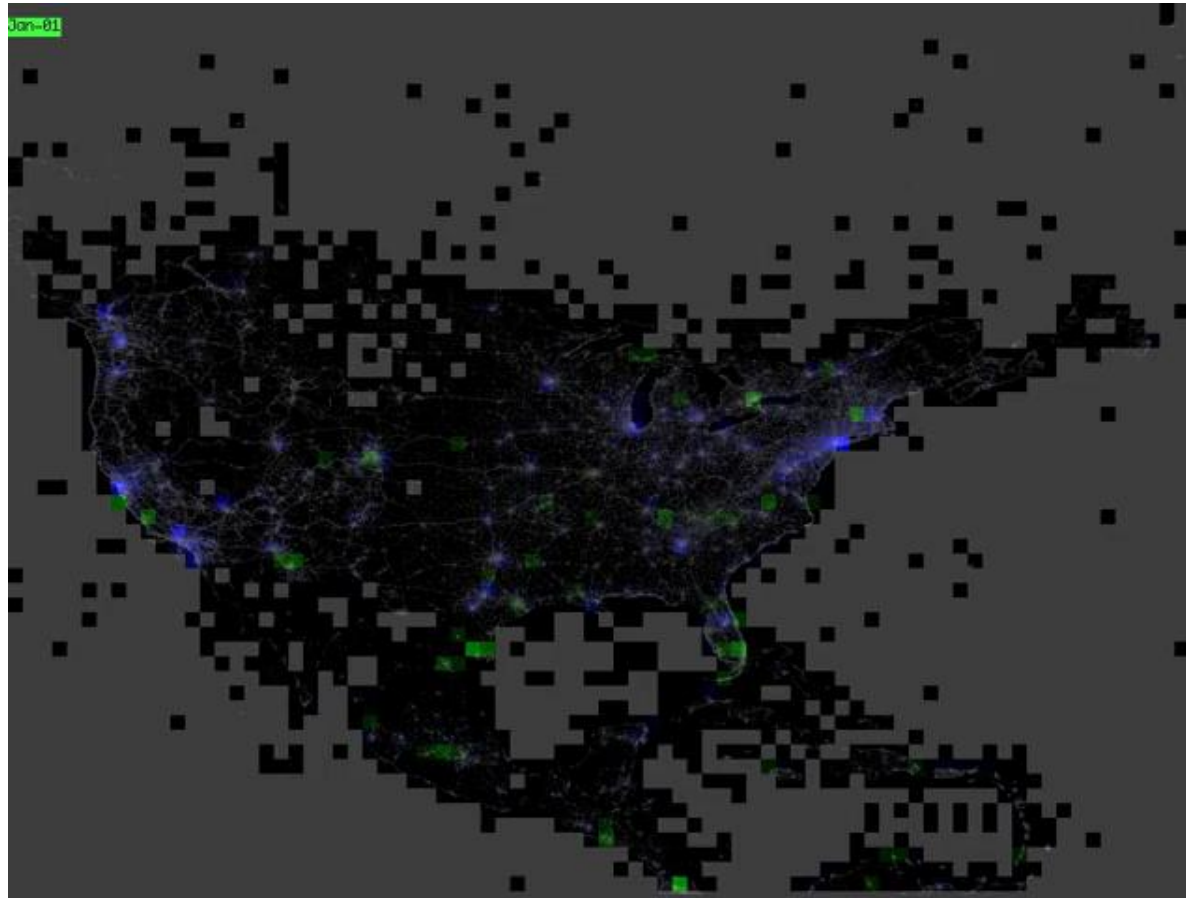


Data's geo property



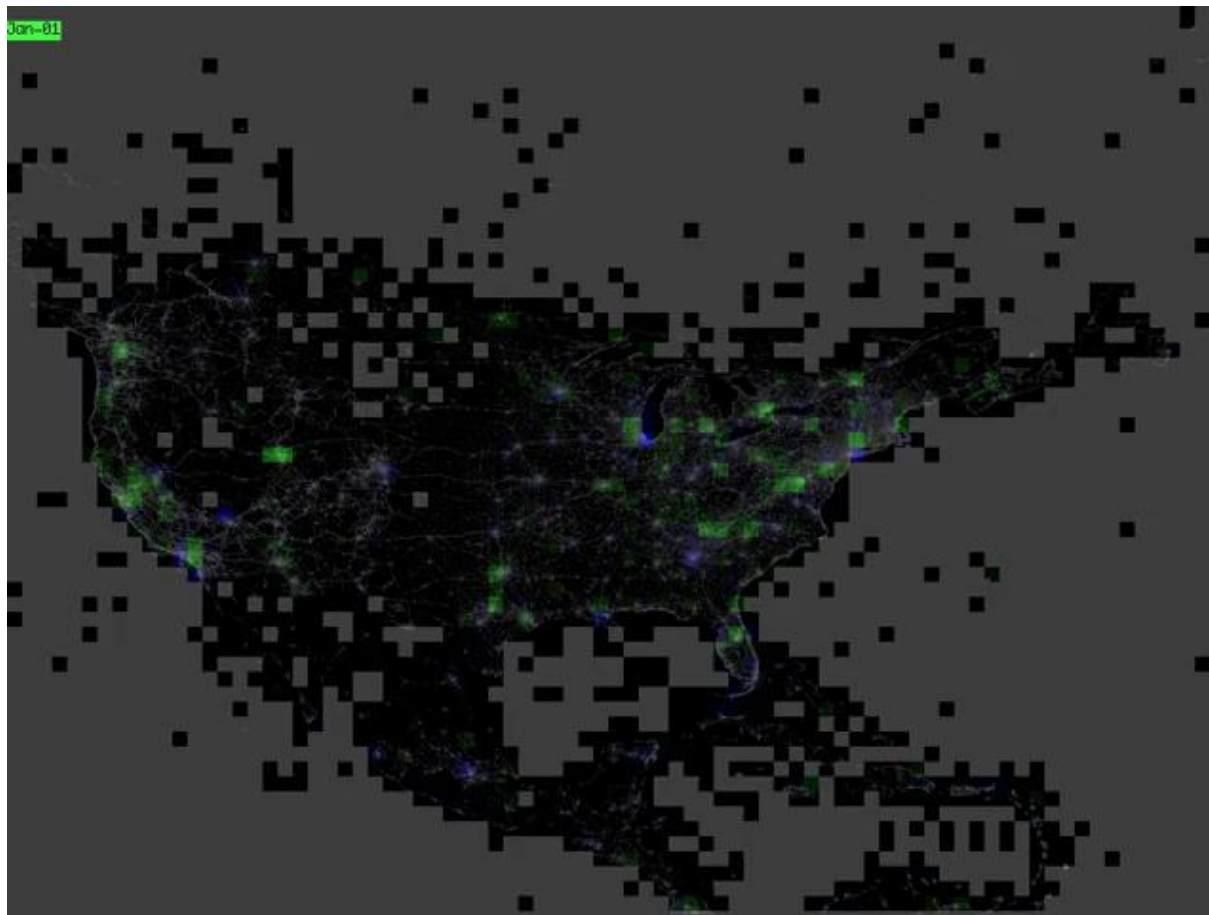
Data's geo-temporal property

Butterfly



Data's geo-temporal property

Leaves

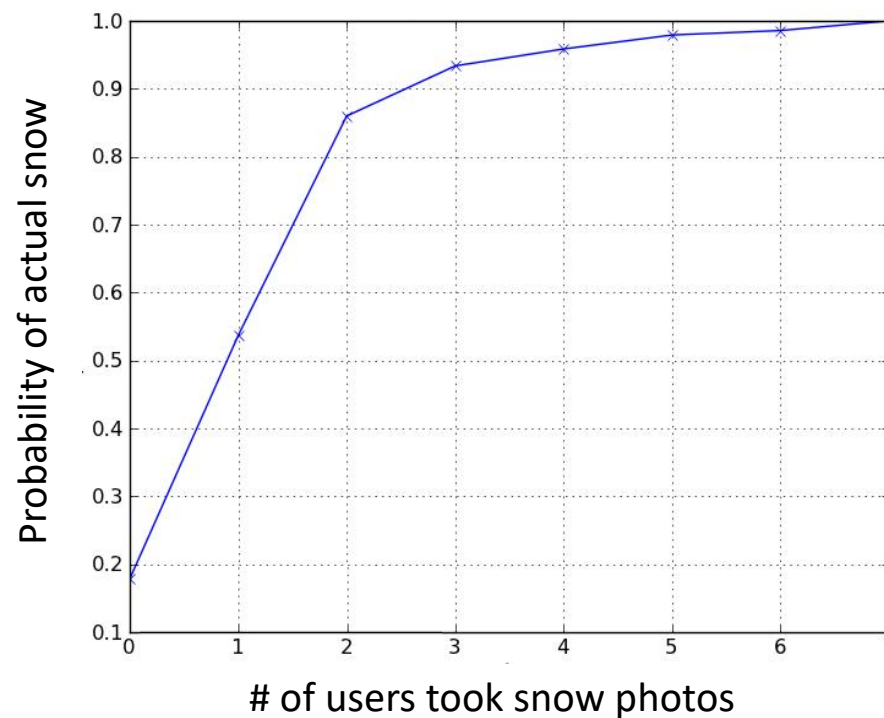


Ecological phenomena detection

- Two phenomena: snow and vegetation cover
 - Estimate geo-temporal distributions at continental scale, using 150M Flickr photos
 - Analyze geo-tags, timestamps, text tags, visual content
 - Evaluate techniques for estimation in crowd-sourced data
 - Compare to weather station and satellite data

Combining evidence

- Individual users have (almost) independent observations, with uncorrelated noise.



Estimate daily snow cover for individual cities

- Estimate parameters on 2007-2008, test on 2009-2010.
- Key statistics for a city on a day:
 - m snow users, # of users took snow photos. (tag set: {snow, snowy...})
 - n no-snow users, # of users did not take snow photos.
- Techniques:
 - Voting: threshold on m .
 - Percentage: $m/(m+n)$.
 - Confidence: likelihood ratio.

m snow users n no-snow users
 $\swarrow$ $\searrow$
 s^m $\bar{s}^n$

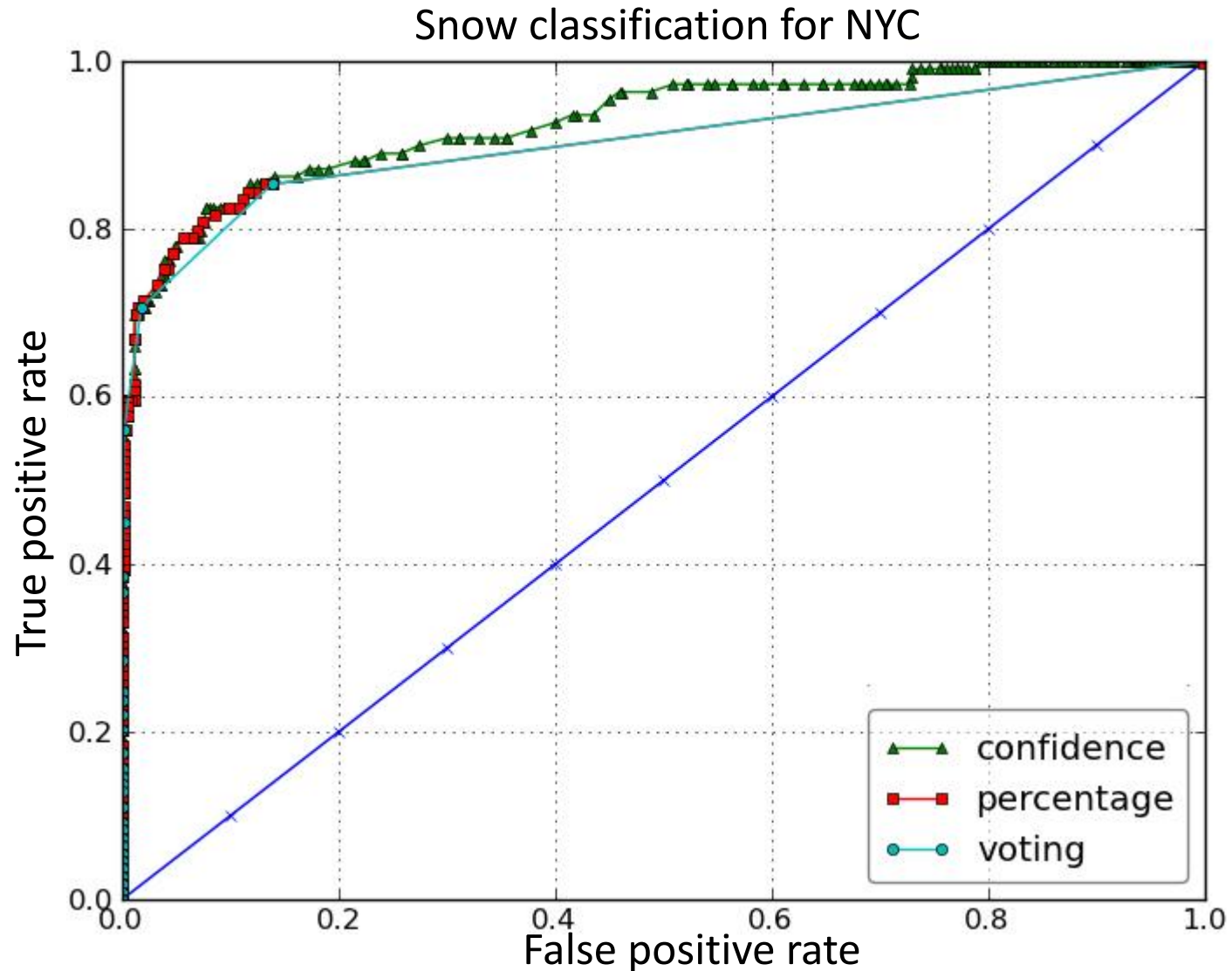
$$\frac{P(\text{snow} | s^m, \bar{s}^n)}{P(\text{snow} | s^m, \bar{s}^n)} = \frac{P(\text{snow})}{P(\text{snow})} \left(\frac{p}{q} \right)^m \left(\frac{1-p}{1-q} \right)^n$$

$P(s | \text{snow})$, prob. of a user taking snow
 photos, given snow was present. 17.12%

 $P(s | \text{no snow})$, prob. of a user taking snow
 photos, given no snow was present. 0.14%

High or low ratio means high or low probability of snow

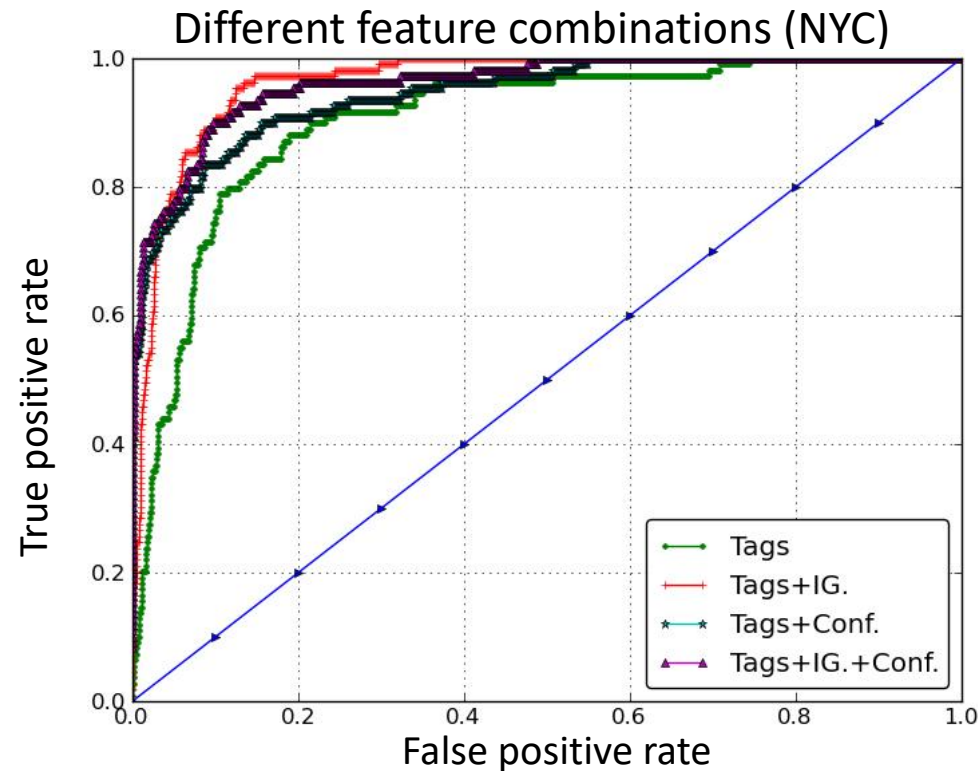
Estimate daily snow cover for individual cities



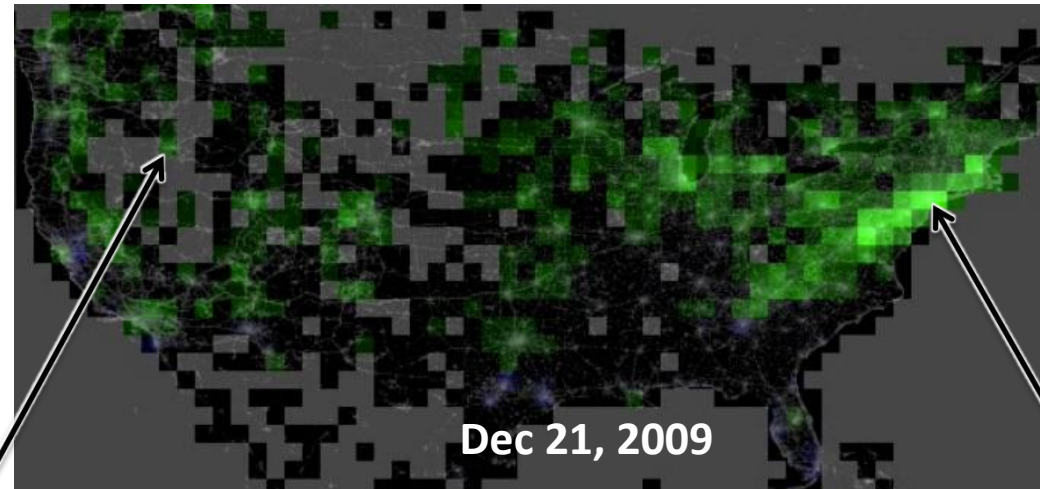
Estimate daily snow cover for individual cities

- SVM classifier:
 - Feature vector: tag usage distribution in a day.
 - Label: snow/no snow from GT.
 - Information gain (IG) to select features.
 - Incorporate confidence score.

Features	Accuracy
Tags	0.859
Tags+Conf.	0.926
Tags+IG	0.91
Tags+IG+Conf.	0.93



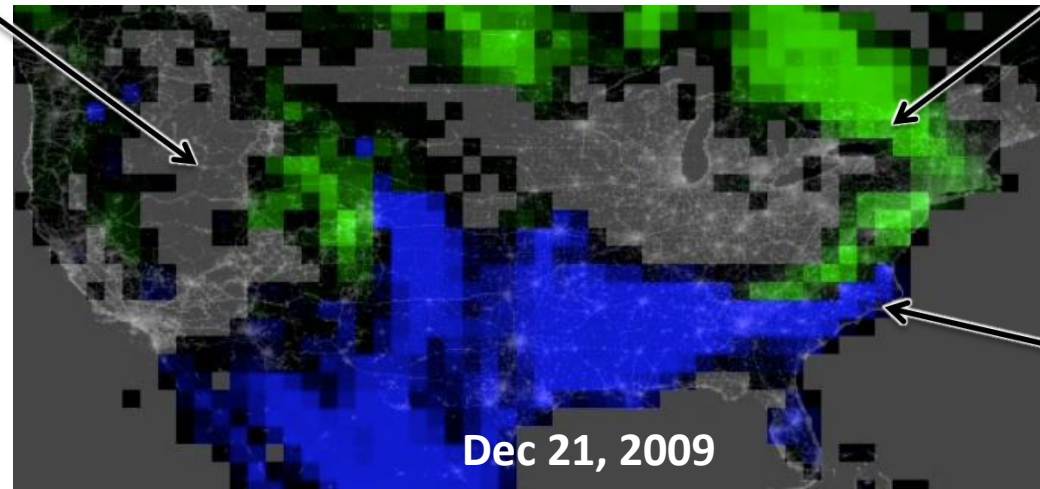
Estimated snow coverage using Flickr photos



Missing
data
(black/grey)

Snow (green)

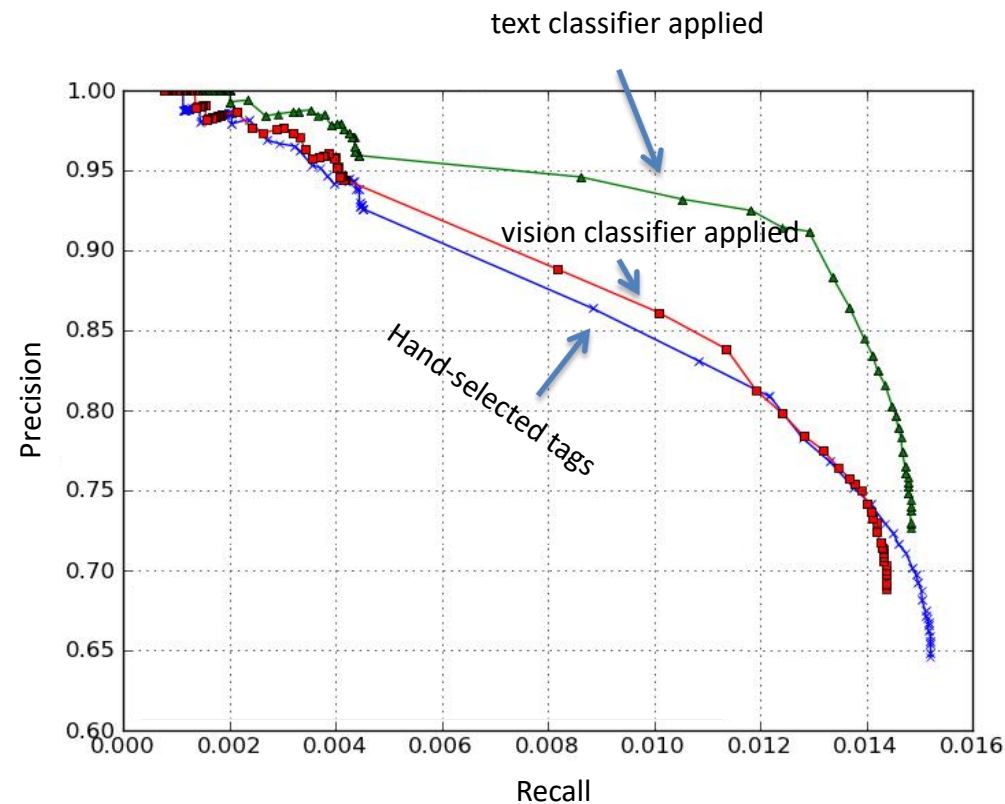
NASA satellite (1 deg*1 deg geobin)



No snow (blue)

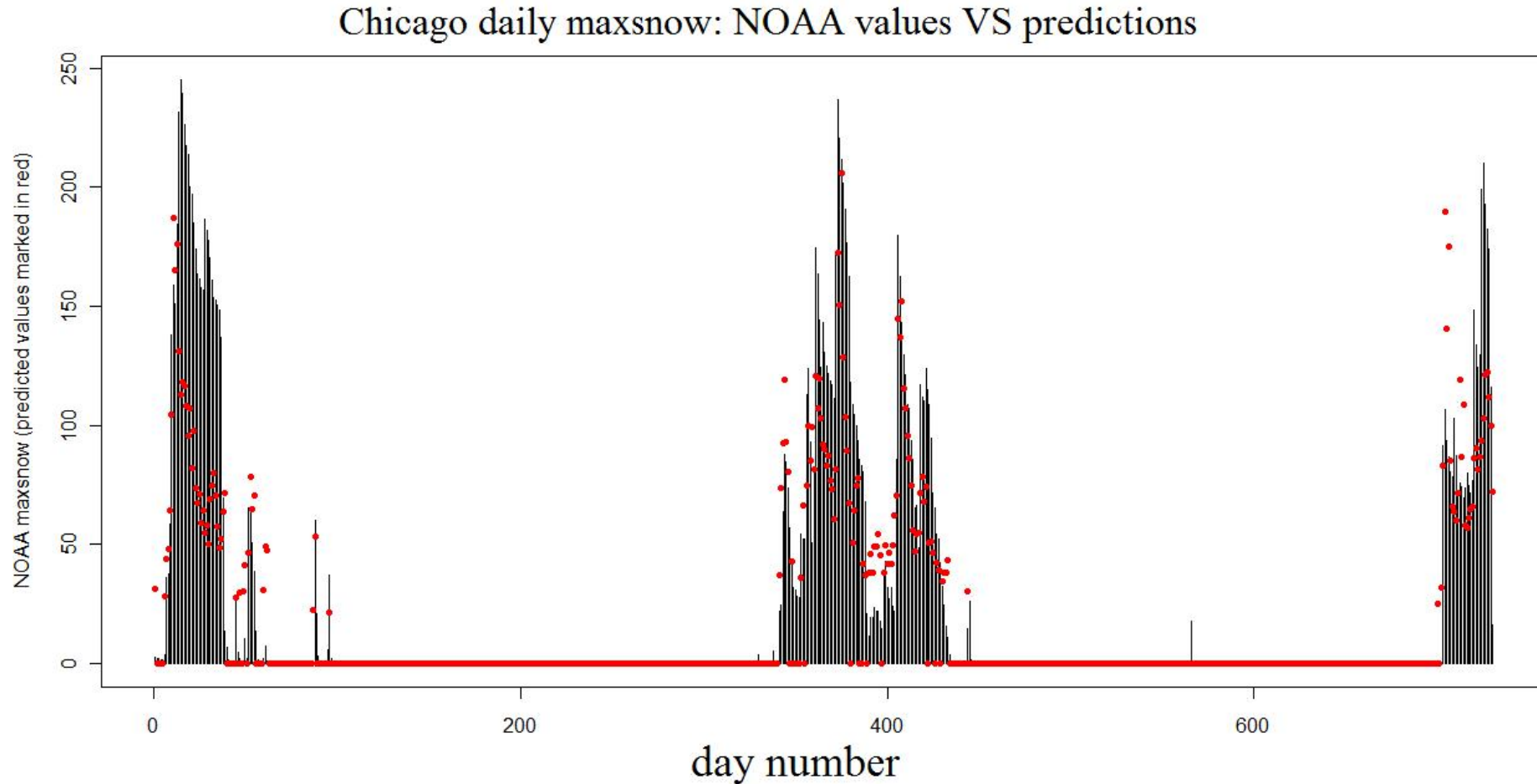
Continental-scale snow detection

- Predict presence of snow, each day for each bin (~35M decisions).
- Vision classifier and textual classifier to filter out false positive photos.



Estimated snow depth

- Chicago daily snow depth (weather stations vs estimations)



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Mining Mobile User Behaviors

Samsung Research America

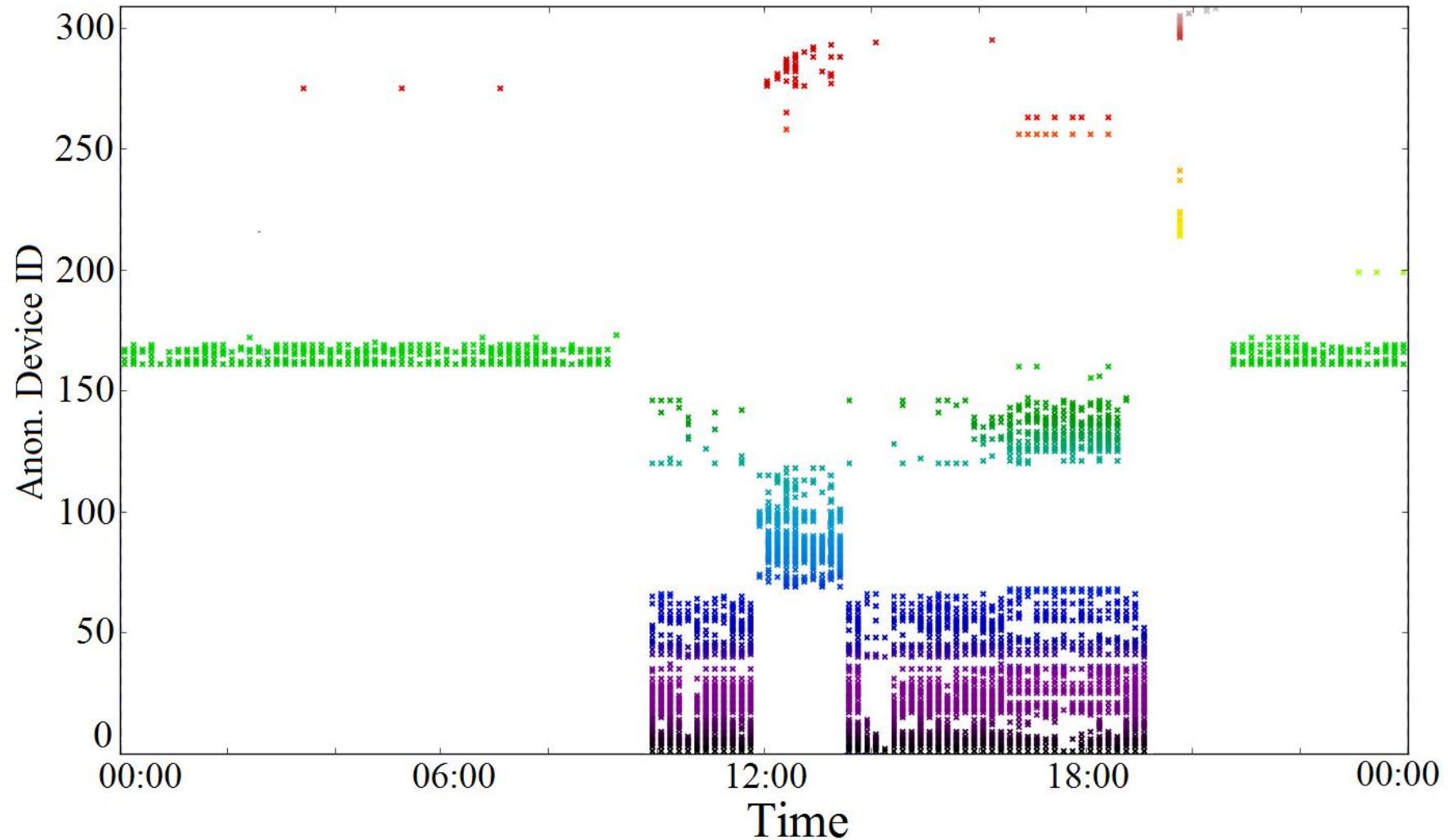
Zhang, H., Yan, Z., Yang, J., Tapia, E. M., and Crandall, D. J. Mfingerprint: Privacy-preserving user modeling with multimodal mobile device footprints. In International Conference on Social Computing, Behavioral-Cultural Modeling, and Prediction (2014), Springer, pp. 195-203. (oral, 24%)

Characterizing users

- Mobile data reveals user characteristics
 - Hobbies
 - Life patterns
 - Work/home locations

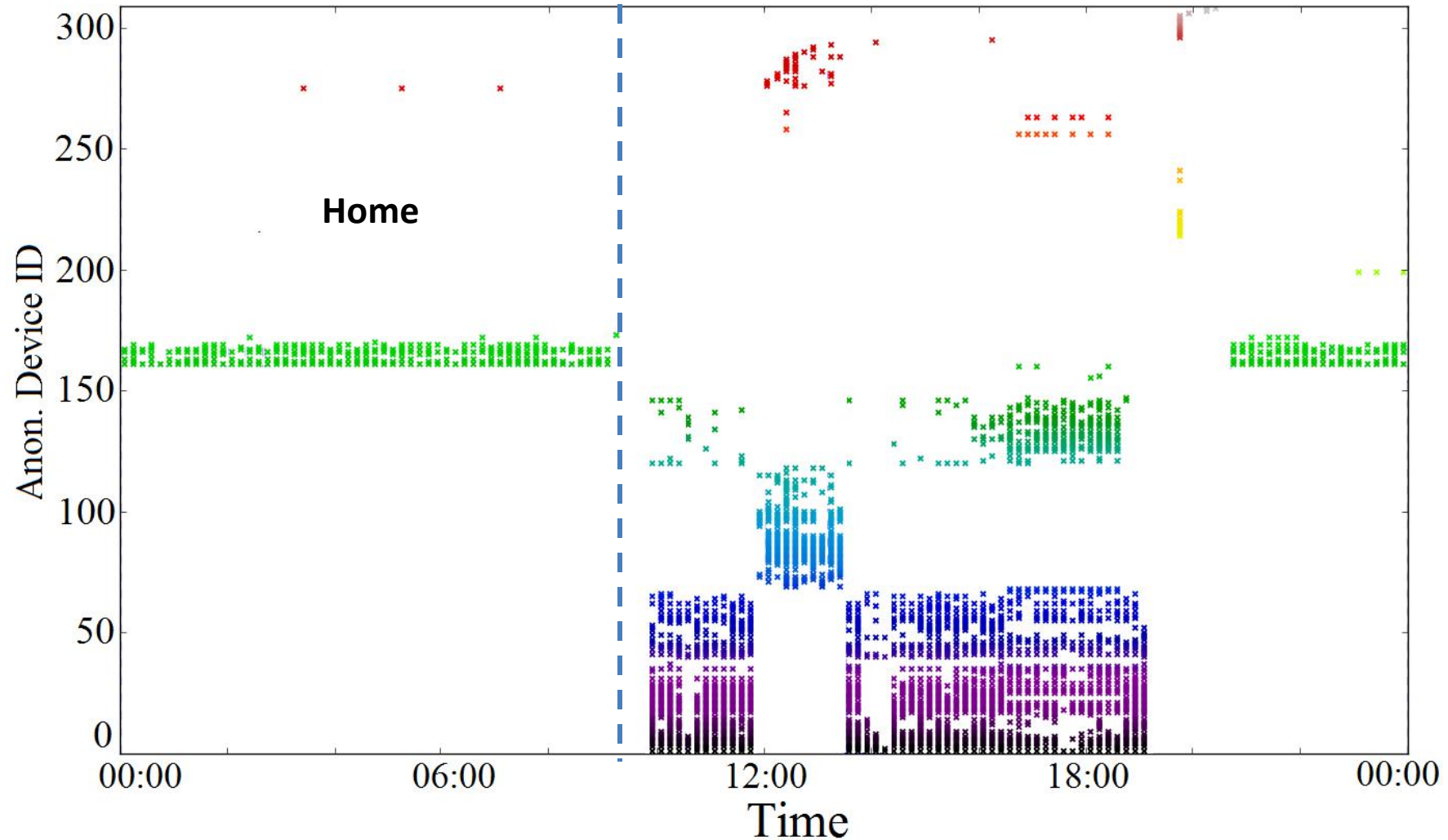
Inferring daily pattern

- WiFi device ID discovered by a phone, in a day



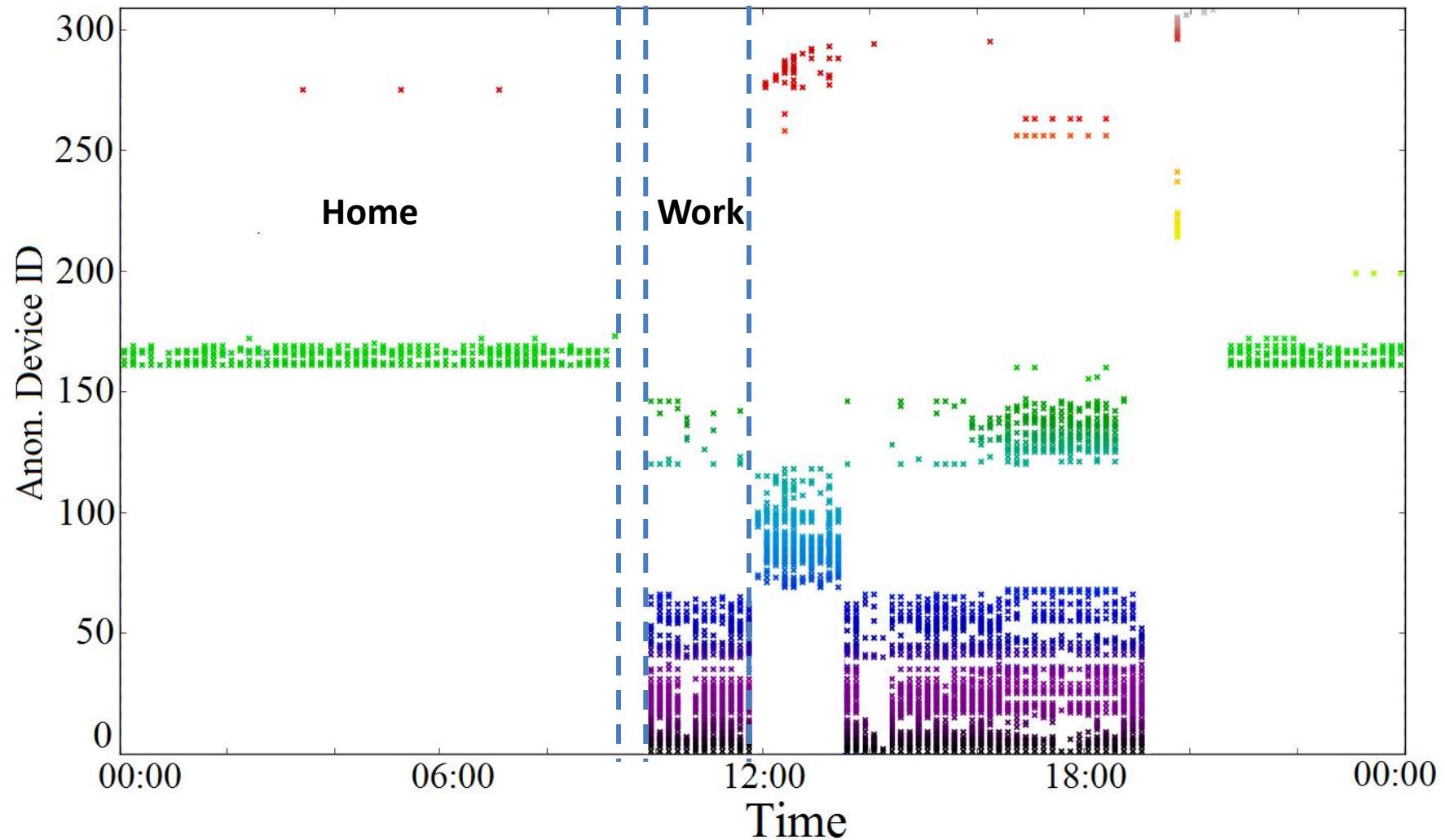
Inferring daily pattern

- WiFi device ID discovered by a phone, in a day



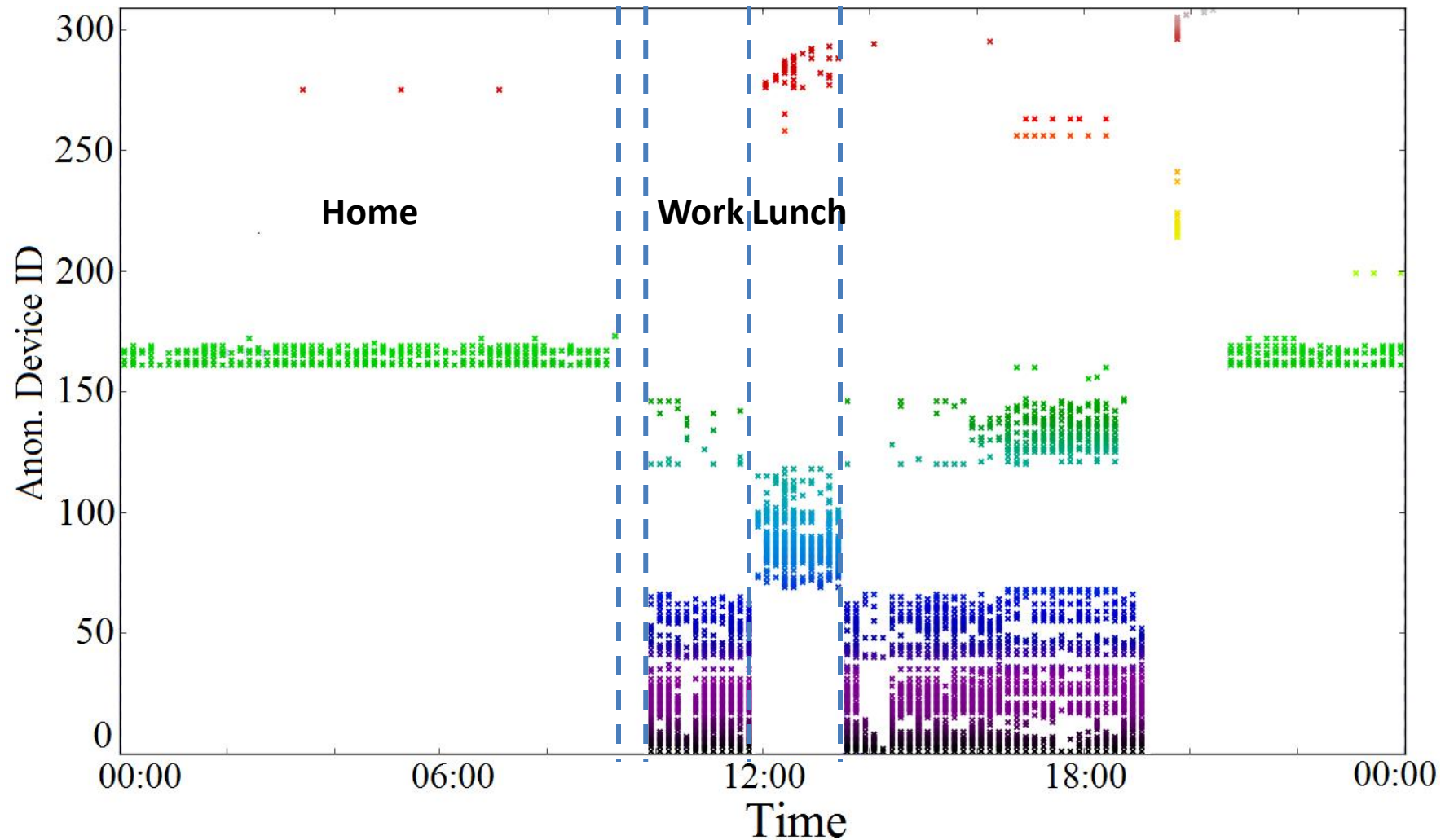
Inferring daily pattern

- WiFi device ID discovered by a phone, in a day



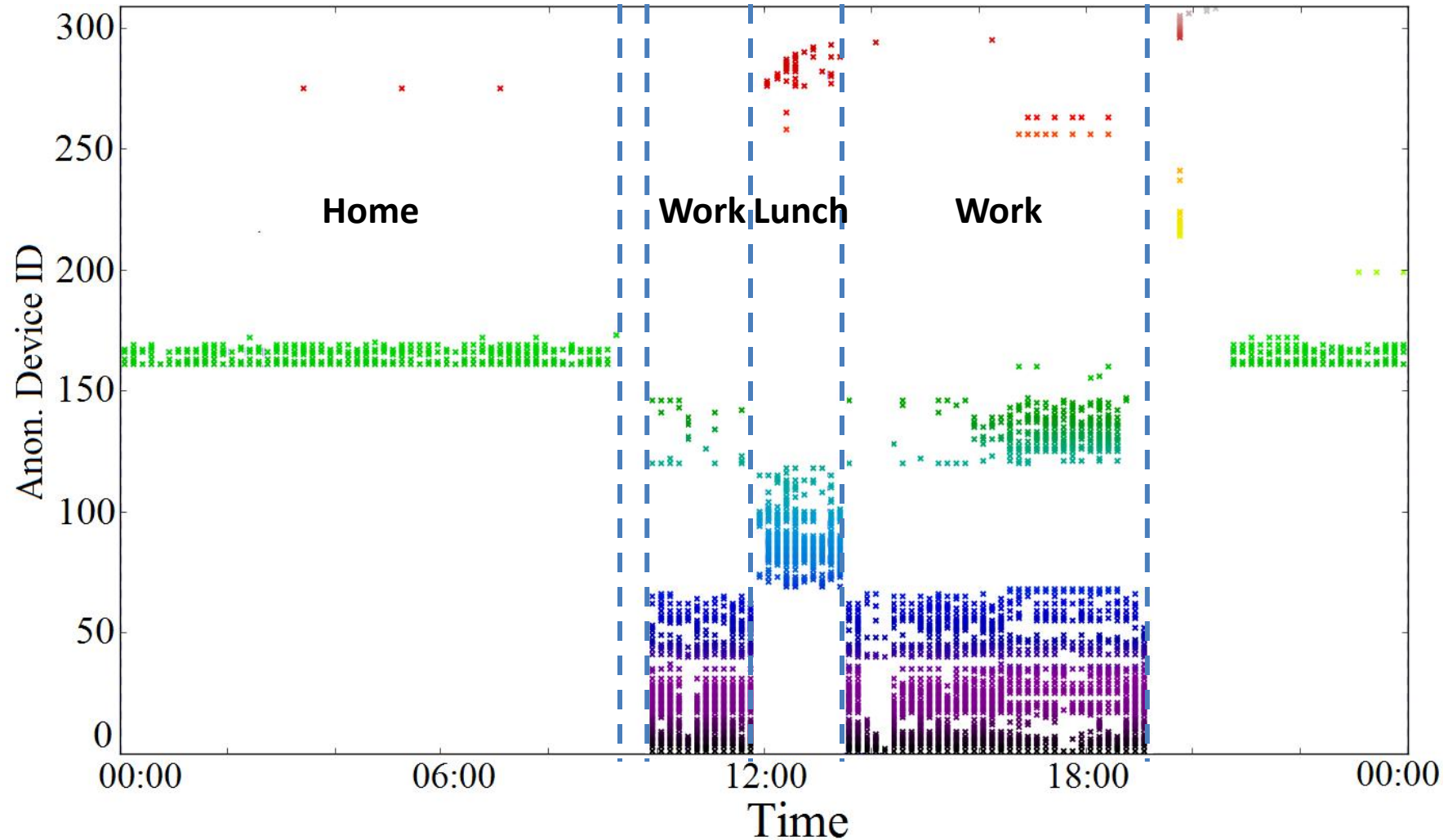
Inferring daily pattern

- WiFi device ID discovered by a phone, in a day



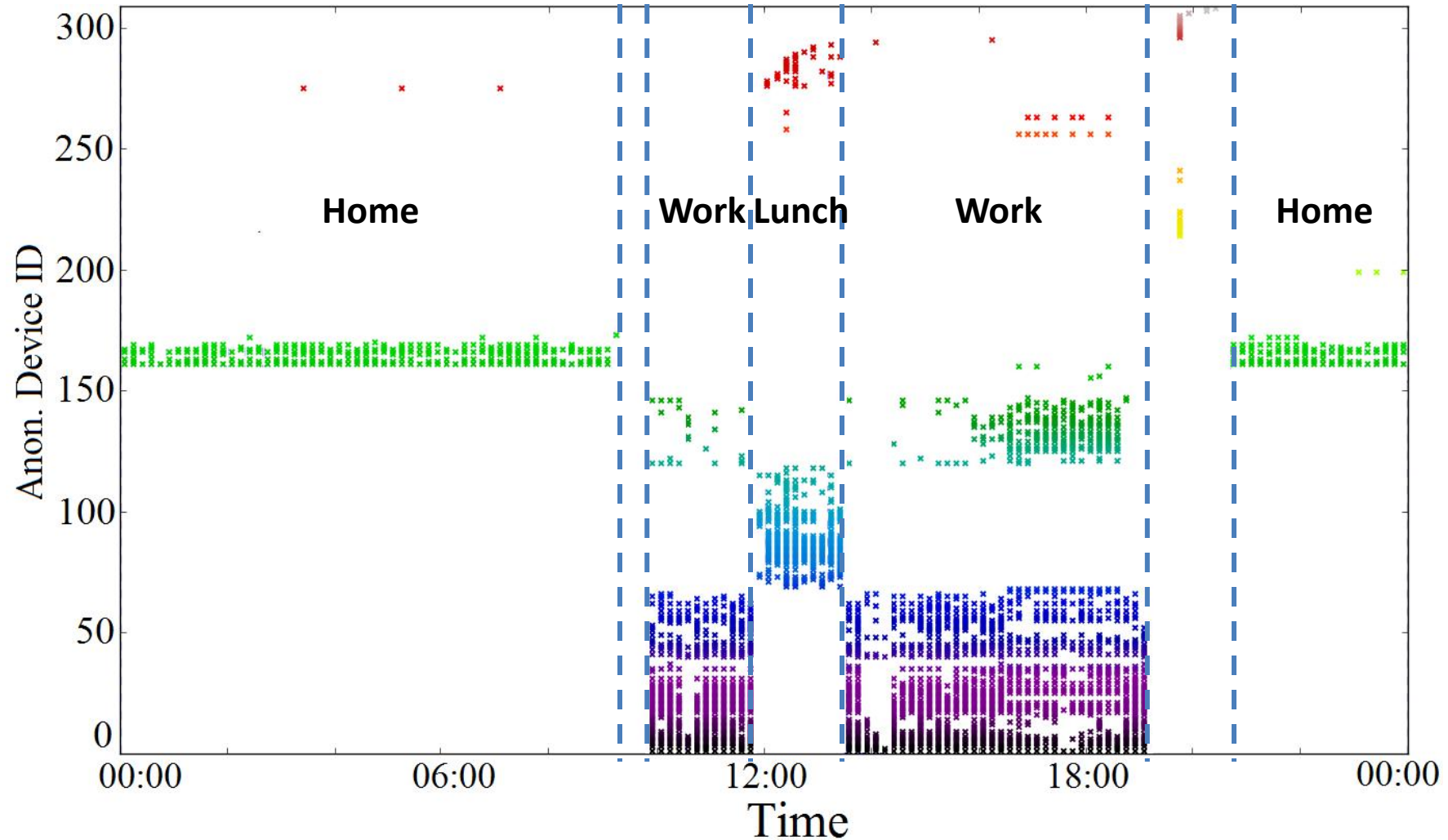
Inferring daily pattern

- WiFi device ID discovered by a phone, in a day



Inferring daily pattern

- WiFi device ID discovered by a phone, in a day



Social media data mining and e-commerce campaign

eBay Research Labs

Zhang, H., Parikh, N., Singh, G., and Sundaresan, N. Chelsea won, and you bought a t-shirt: Characterizing the interplay between twitter and e-commerce.
In Proceedings of the 2013 IEEE/ACM international conference on advances in social networks analysis and mining (2013), ACM, pp. 829-836. (oral, Best Paper Award)

Social media & consumer behavior

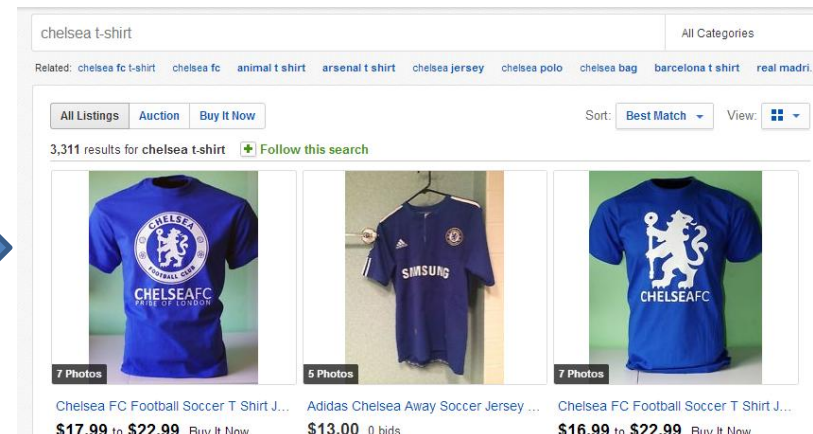
- Social media and e-commerce are usually studied independently.
 - e-commerce trending search queries: [Parikh08]
 - last-minute bidding: [Roth02] seller reputation: [Houser06]
- Bridge the two fields:
 - Temporal relationships between user behaviors on eBay and Twitter
 - Predict trends on eBay from signals on Twitter and build recommender systems.

Twitter

Retweeted by Chelsea FC
André Schürrle @Andre_Schuerlle · Aug 18
Great start in the saison...come on blues!!! #cfc

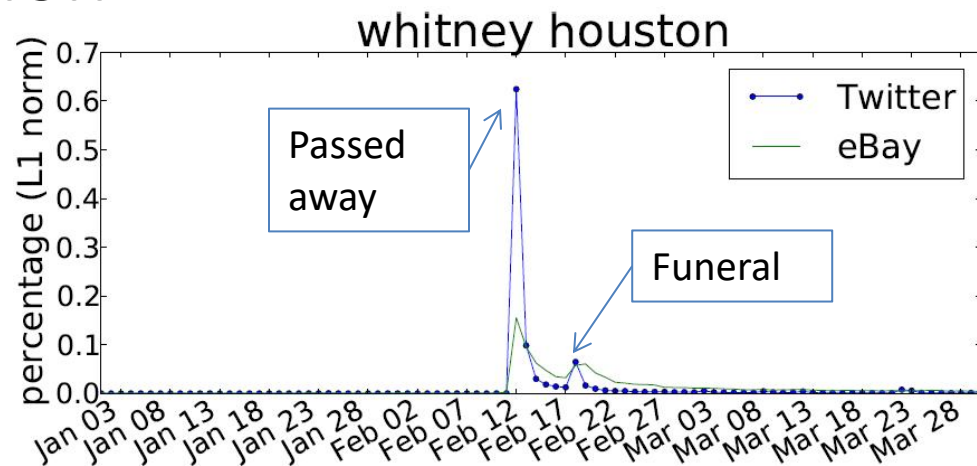


eBay

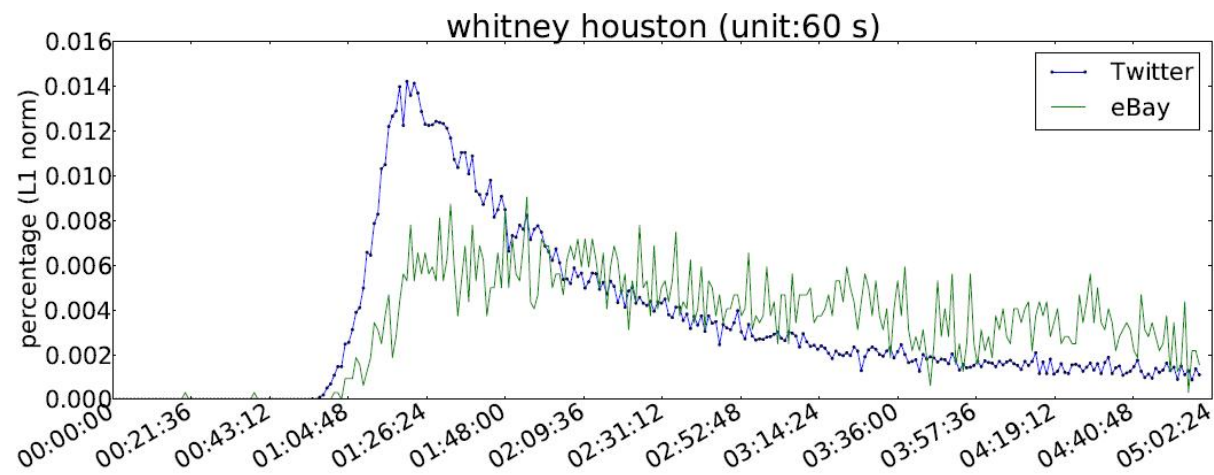


Extract time series to observe user behaviors

- Correlation



- Lags



Extraction of time series

- Temporal trends of keyword **k** on eBay and Twitter.
- Divide the period of time into temporal bins.

Trend name		descriptions
Twitter mention		Each daily bin records # of unique users whose tweets contain k
eBay	SEARCH	# of unique users with queries containing k
	VIEW	# of clicks on resulting items
	BID	# of bids on results
	BIN count	# of purchased BIN items
	total BIN price	total prices of purchased BIN items
	Avg. BIN price	Avg. price of purchased BIN items

*BIN: Buy It Now, fixed-price items

Quantifying correlations

- Pearson correlation measures the linear dependence between two datasets, gives a co-efficient r in $[-1,1]$.

$$r = P(E, T) = \frac{\text{cov}(E, T)}{\sigma_E \sigma_T}$$

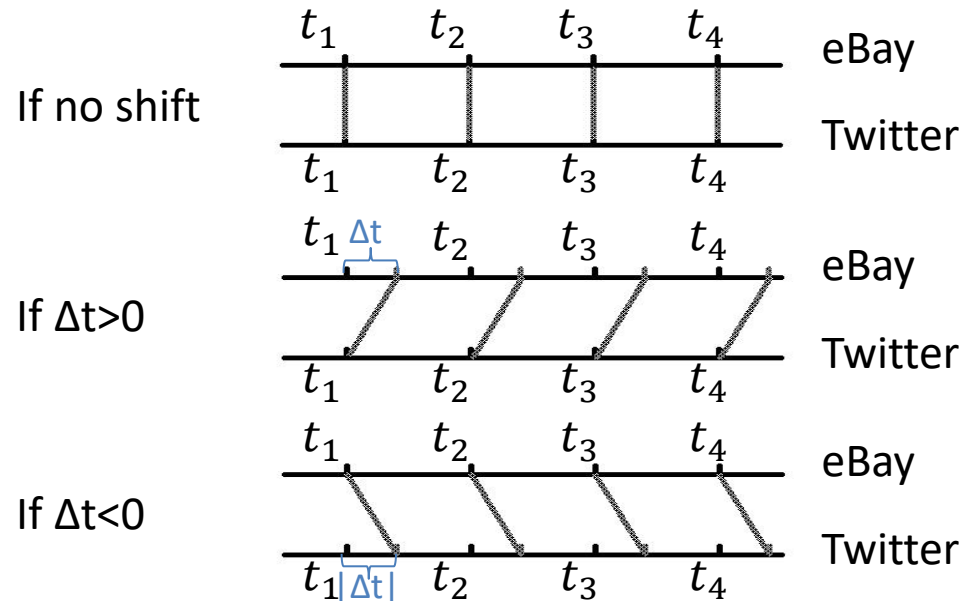
E: eBay vector; T: Twitter vector

- Student's t-test on r . Null hypothesis: E and T are uncorrelated (in this case r has a t-distribution)*.

*It holds if the observed values are from normal distributions. Even if they are non-normal, it approximately holds when sample sizes are not small. [Kendall74]

Computing lags

- A moving window approach: for each pair of streams, shift one stream by a period of time Δt to compute r .



- Lag ℓ is the Δt that maximizes Pearson's r (Δt in $[-T, T]$). Positive ℓ suggests eBay lags Twitter by ℓ .

Experiments (correlations)

- Top 150K queries on eBay, for three months.
- **GeneralQueryList**: threshold on daily count → 16K queries.

Fractions of correlated pairs at different confidence levels

p-value	pos_corr	neg_corr	pos/neg
0.01	8.52%	2.49%	3.42
0.005	7.34%	1.77%	4.14
0.001	5.75%	0.80%	7.18
0.0005	5.25%	0.65%	8.07
0.0001	4.35%	0.32%	13.59

Top eBay Categories ranked by query's correlation rate (p=0.0005)

Query Category	pos_corr
Video Games	21.28%
DVDs & Movies	14.20%
Entertainment Memorabilia	13.47%
Sports Mem, Cards & Fan Shop	13.45%
Tickets	13.04%

Experiments (correlations)

- **TrendQueryList** : trending queries [Parikh08] from GeneralQueryList → 730 queries

Fractions of correlated pairs at different confidence levels

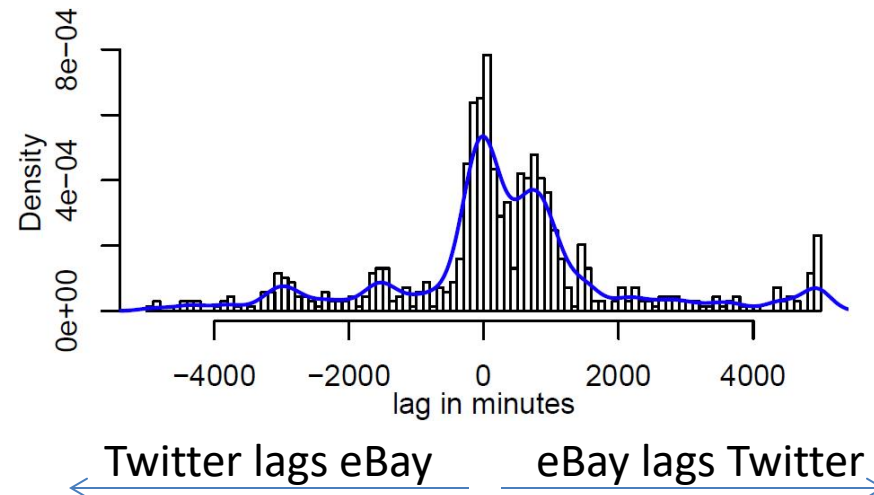
p-value	pos_corr	neg_corr	pos/neg
0.01	29.58%	1.64%	18.03
0.005	28.21%	1.09%	25.88
0.001	25.61%	0.13%	197
0.0005	24.93%	0.13%	191.76
0.0001	22.46%	0%	INF

Top eBay Categories ranked by query's correlation rate (p=0.0005)

Query Category	pos_corr
Sports Mem, Cards & Fan Shop	69.23%
Video Games & Movies	64.28%
Cell Phones & PDAs	52.94%
Entertainment Memorabilia	37.50%
Music	36.66%

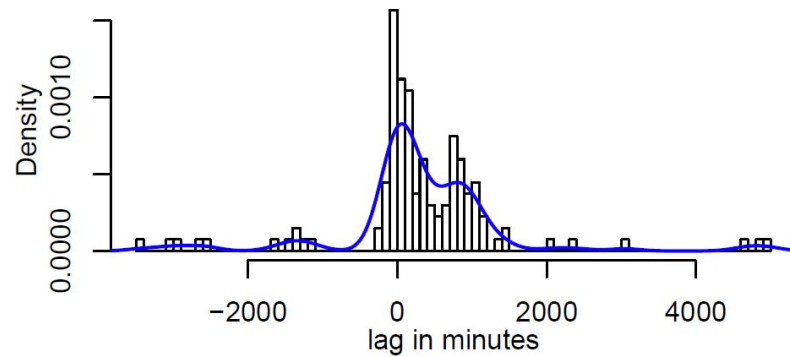
Experiments (lags)

- **CorrelatedQueryList**: a list of strongly correlated pairs ($r \geq 0.4$) from GeneralQueryList $\rightarrow$ 690 queries.
- Shift range: [-5000, 5000] minutes.
- Avg. lag value 4.8 hours; 61.30% have positive lags.

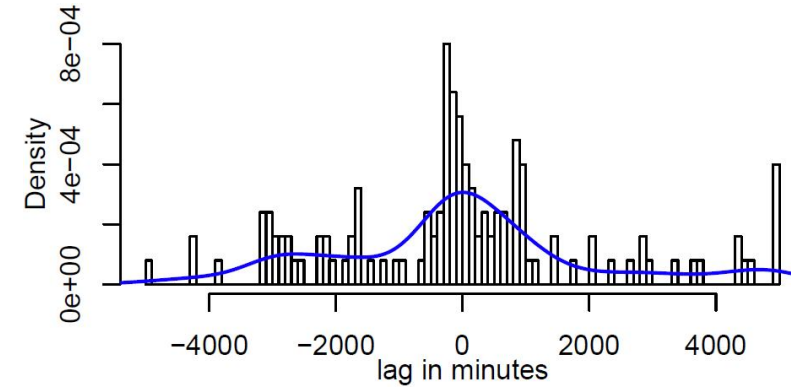


Experiments (lags)

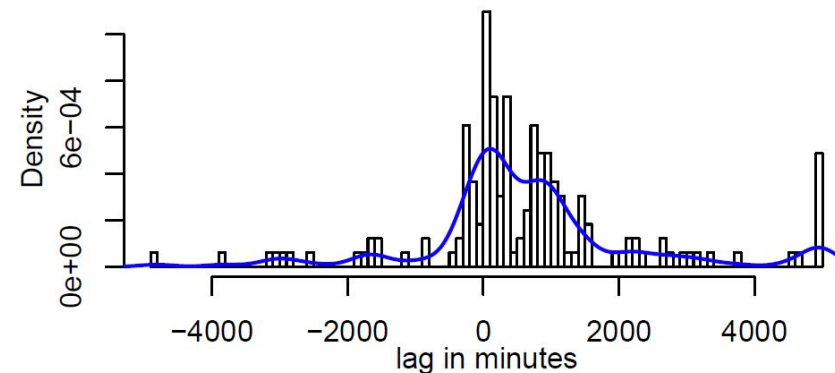
'Sports' from CorrelatedQueryList, 74% have positive lag values



'Clothing' from CorrelatedQueryList, 45.6% have positive lag values

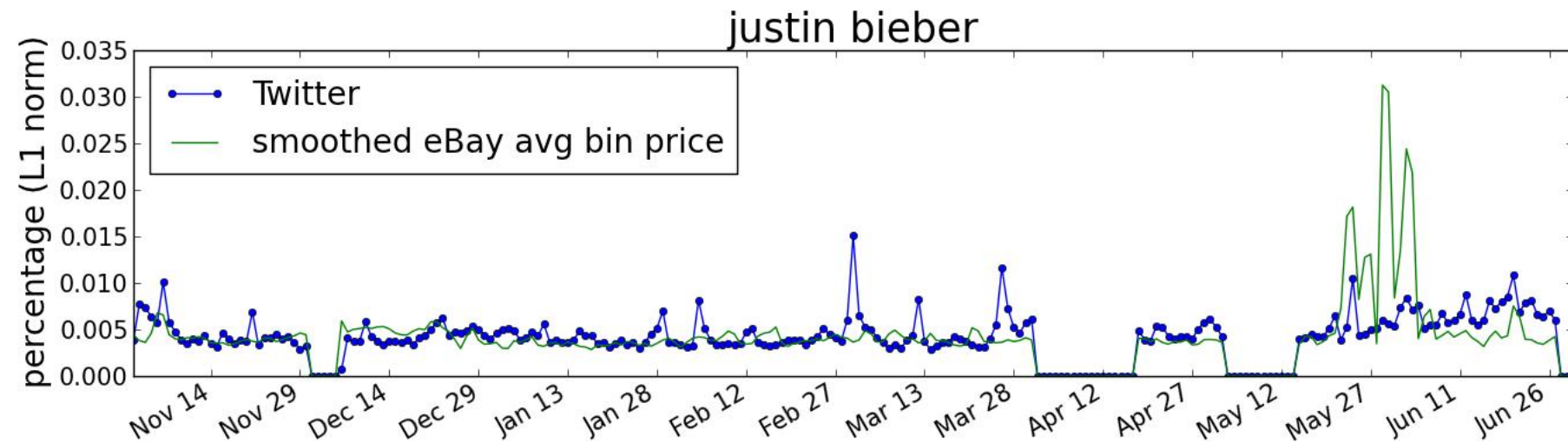


- Trending queries from CorrelatedQueryList → 164 queries, 76.82% with positive lags. Avg. lag: 11 hours.



Other findings

- Celebrities' Twitter popularities correlate with eBay search and sale trends, impact avg. prices.



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 - 挖掘卫星图片预测经济指标
 - 突发风险事件监测与市场建模

挖掘卫星图片预测经济指标

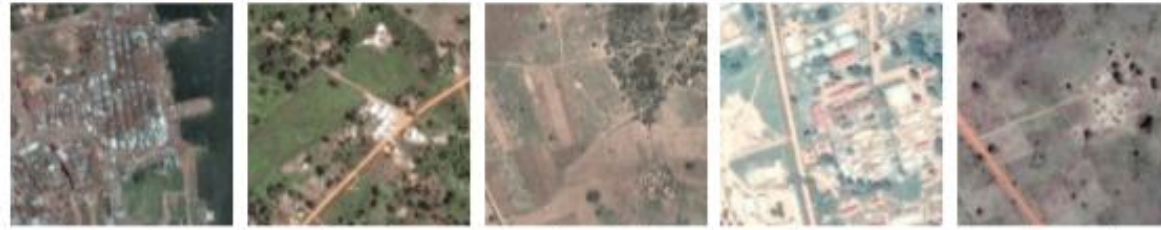
Jean, Neal, et al. "Combining satellite imagery and machine learning to predict poverty." **Science** 353.6301 (2016): 790-794.

课题背景

- 国际贫困线: 每天1.90美元
- 全球绝对贫困人口10亿
- 对于欠发达国家而言, **缺少可靠的贫困相关数据**, 难以有的放矢制定扶贫政策

- 目标：仅使用日间卫星图片等公开数据，准确而全面地预测贫困指标

Inputs: Daytime satellite imagery



X_1 X_2 X_3 X_4 ... X_n

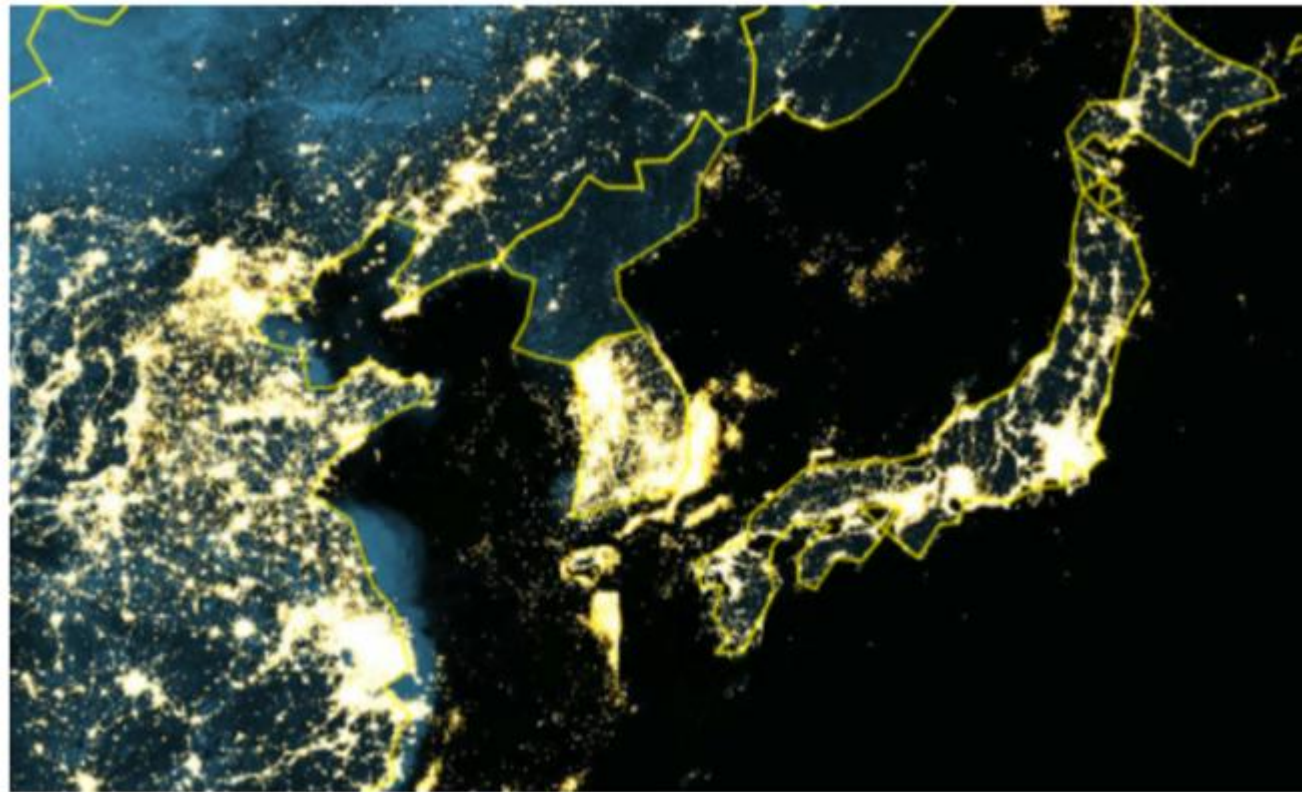
Convolutional Neural Network

Low-dimensional feature representation

Y_1 Y_2 Y_3 Y_4 ... Y_n

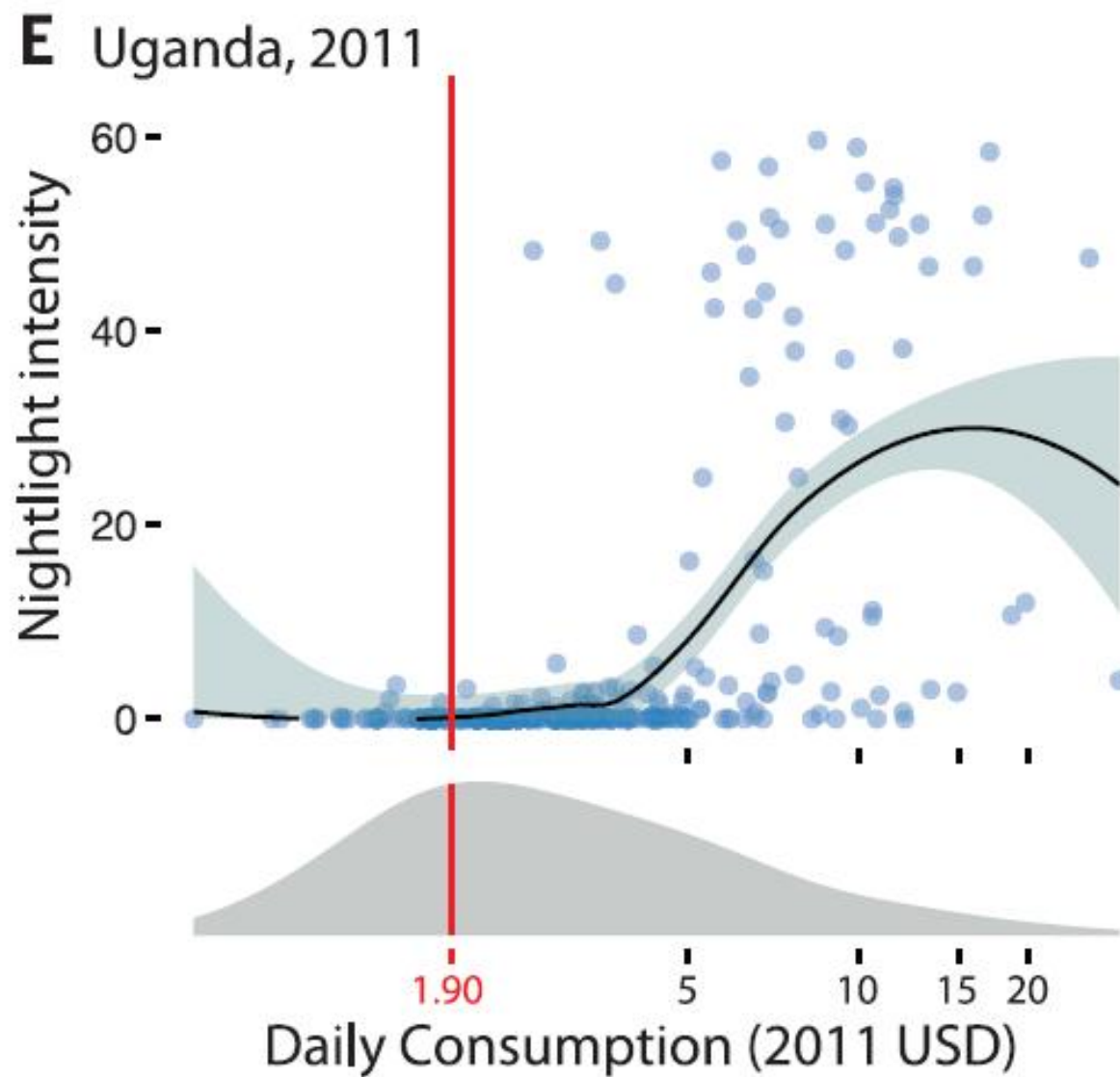
Predictions: Economic indicators

- 常规的有监督学习方法无法使用：缺少足够的ground truth (真实值)
- 核心想法： **使用夜晚光照强度作为经济发展水平的代理指标**

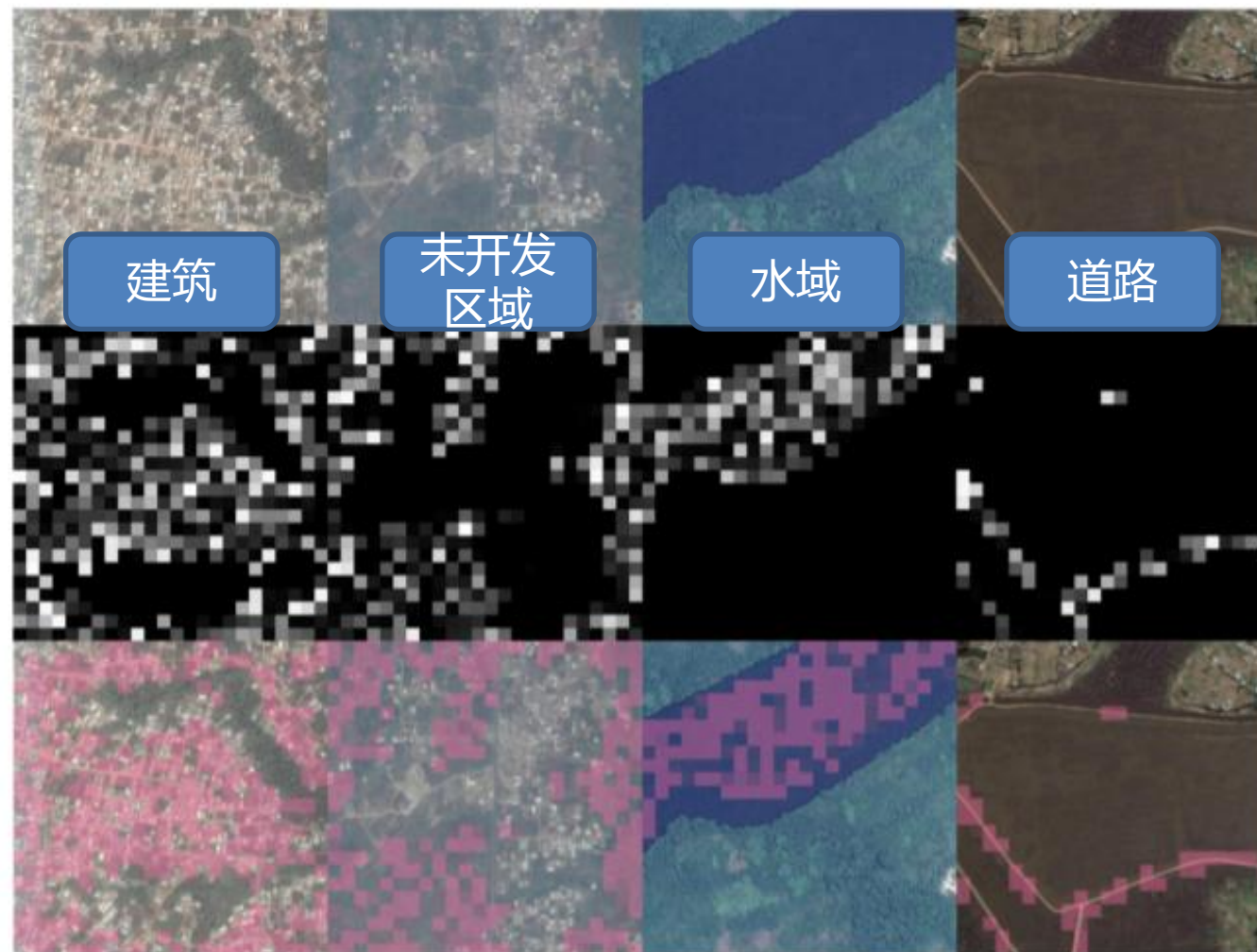


North Korea vs. South Korea

夜晚光照强度无法直接估计贫困指标



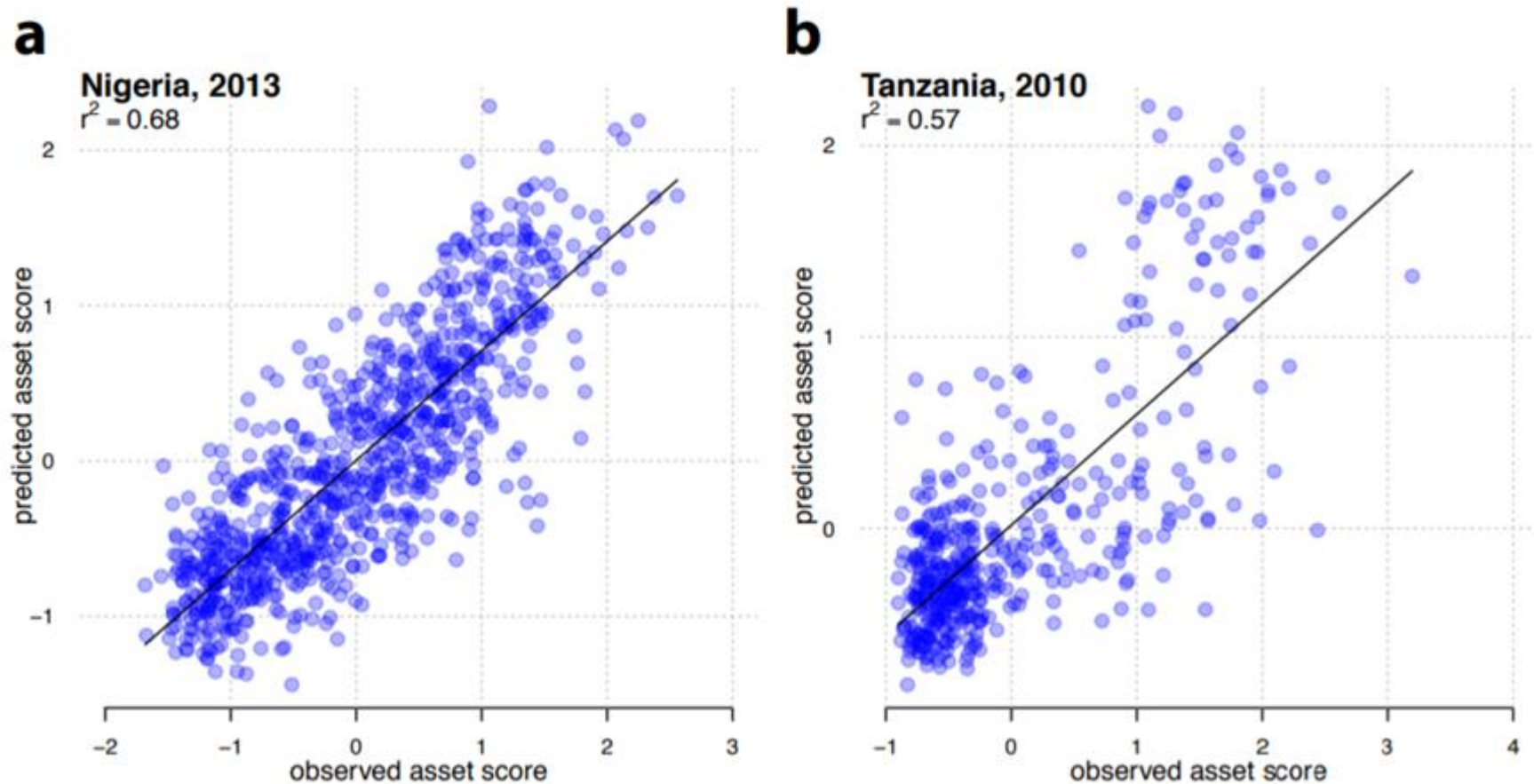
学习到的图像特征



Learned features corresponding to buildings, undeveloped areas, water, and roads

用图像特征与经济指标建立预测模型

预测非洲5国的消费与资产指标



Asset-based wealth predictions for (a) Nigeria and (b) Tanzania

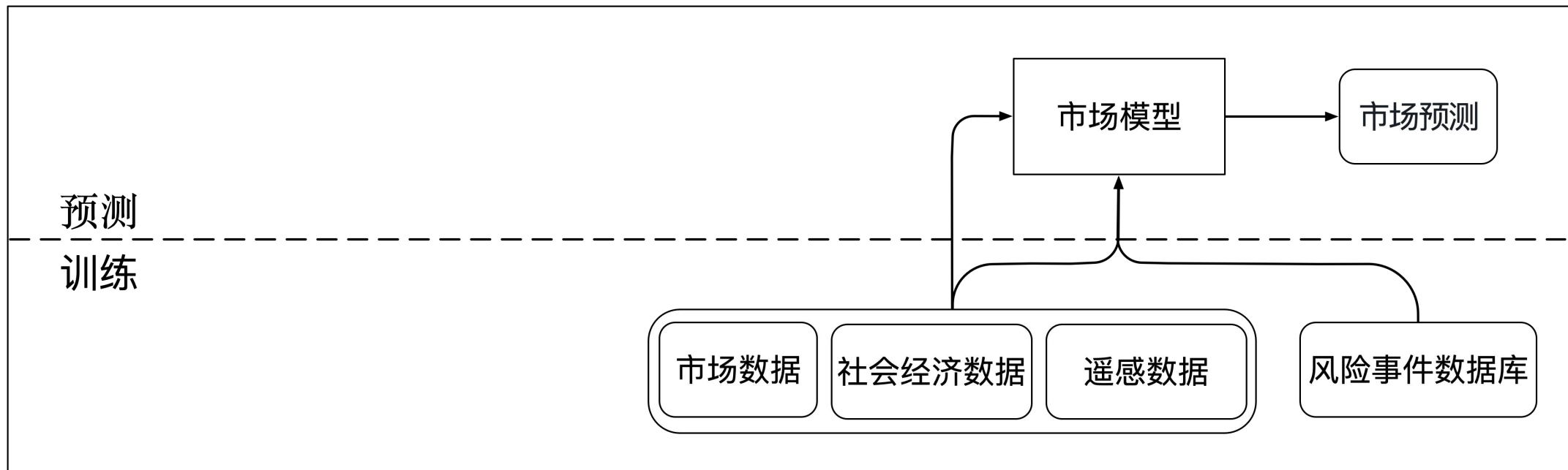
突发风险事件监测与市场建模

One Step Ahead: A Framework for Detecting Unexpected Incidents and Predicting the Stock Markets

课题背景

- 各类突发风险事件（黑天鹅）冲击市场
 - 911事件（2001），道琼斯五日 -17.5%
 - 波士顿爆炸（2013），标普500 -2.3%
 - 伦敦爆炸案（2017），富时100 -4%
 - 福岛核泄露（2011），日经两日 -15.6%
 - 2000年以来，全球年均6580起恐袭、178次6级以上地震、105次飓风、10次海啸
- 研究问题
 - 能否实时**发现风险事件并提取主要信息？**
 - 能否实时**预测事件对市场的影响？**

框架与方法



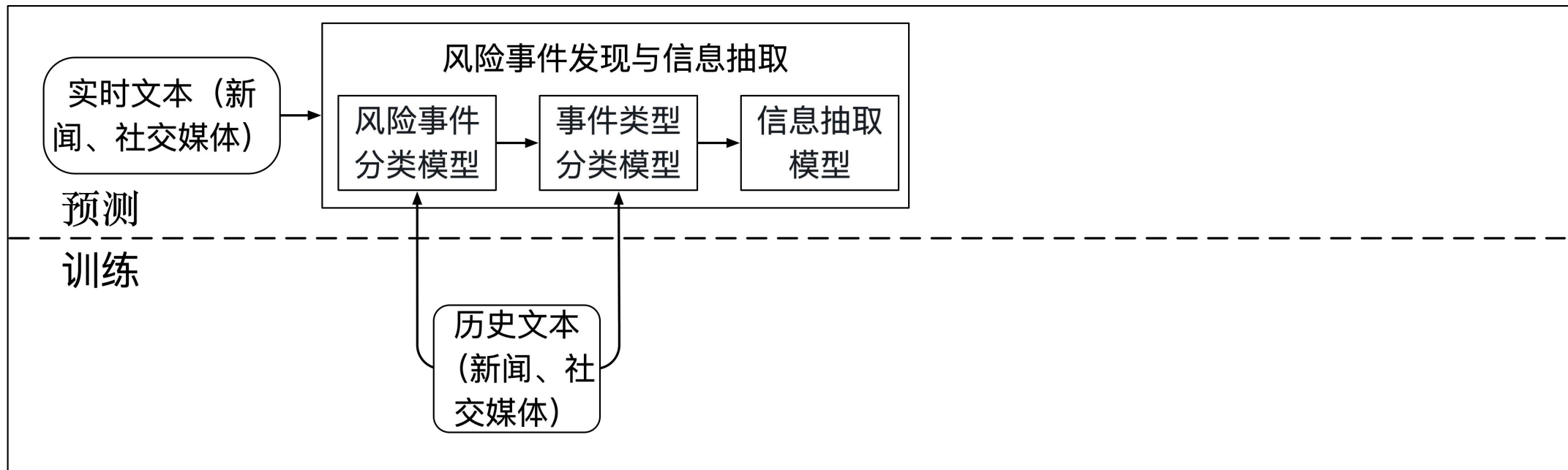
- 市场建模

- 基于历史事件数据库、市场数据、政治经济数据集
- 机器学习模型：决策树、SVM、逻辑回归、随机森林

- 实时事件发现

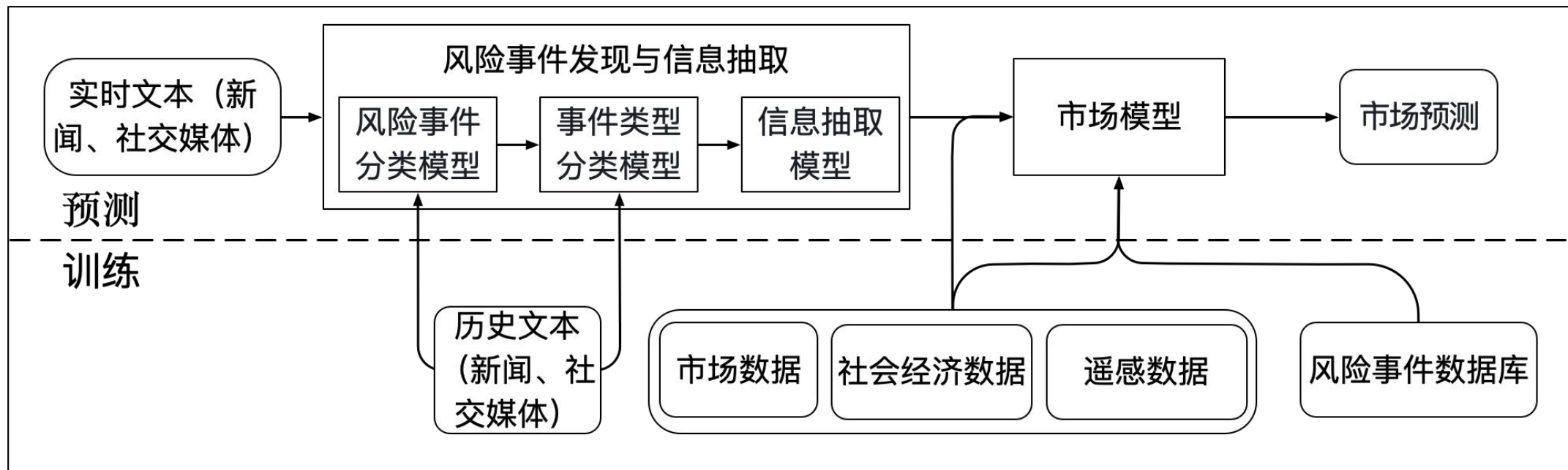
- 新闻分类：恐怖袭击、自然灾害、传染病3大类12个小类 (基于CNN的分类)
- 信息抽取：时间、地点、伤亡人数、武器类型、灾害等级... (基于句法依存的抽取)

框架与方法



- 市场建模
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夜间光照卫星图片

- 多角度描摹事件
 - 发生在经济发达地区与欠发达地区事件影响不同
 - 客观、全覆盖、可比较



实验设计

- 对**恐怖袭击与市场**建模
 - 7:3划分训练与测试集

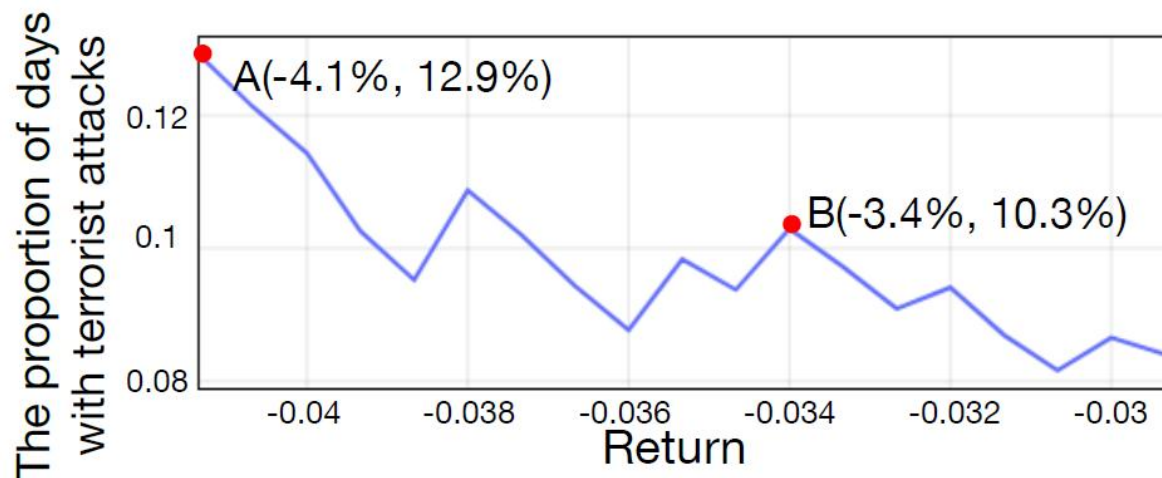
市场数据	时间	恐袭次数	恐袭数据库 (GTD)
以色列 (TA100)	25年	1,224	共135个恐怖袭击事件相关特征, 包括死伤人数、发生地点经纬度、攻击目标等
哥伦比亚 (IGBC)	14年	1,427	
西班牙 (IGBM)	23年	680	

- 模拟实时新闻, 进行事件发现与抽取

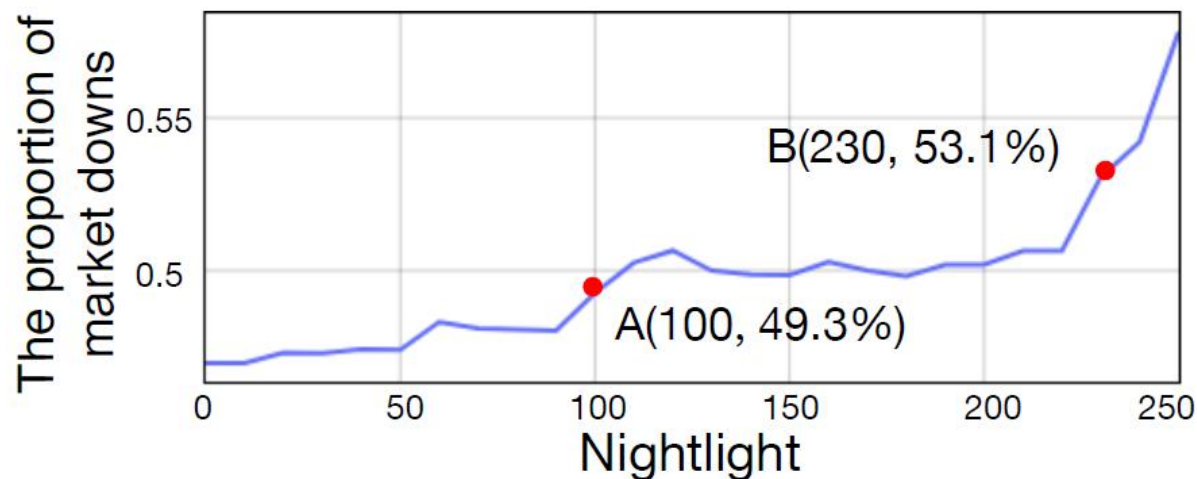
新闻数据	数量	属性
路透社	9,529,834	发布时间, 标题
Wikipedia	13,644	标题, 事件类型

发现

- 恐怖袭击对市场存在影响：市场下跌越多的日子发生恐袭的比例越高



- 夜光越强（经济越发达）的地方发生恐袭，越可能下跌



实验结果

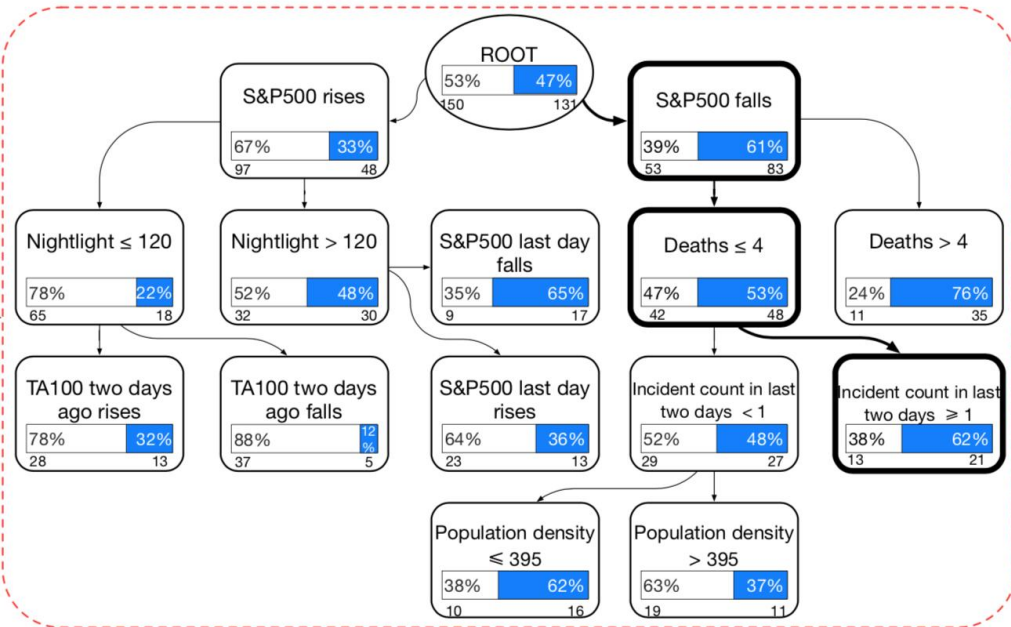
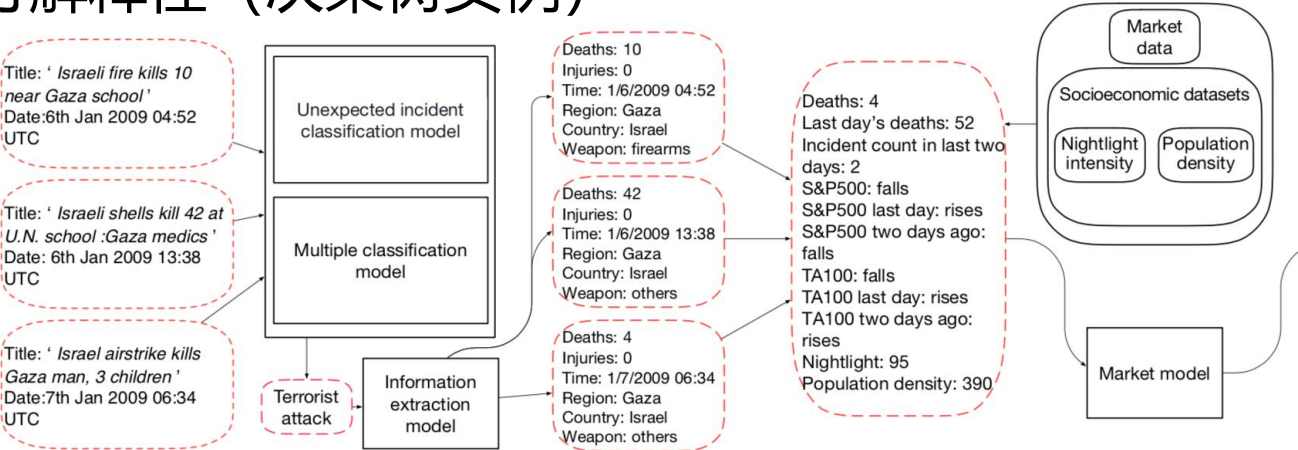
事件发现与信息抽取

	恐怖袭击	其他类型
分类precision	92.6%	84.0%
抽取accuracy	88.0%	74.0%

市场预测

	全特征		基线（仅市场）	
	准确率	召回率	准确率	召回率
三国加总	68.3%	55.5%	66.3%	50.8%

可解释性（决策树实例）



另类数据挖掘及其在金融中的应用

- **另类数据及其商业化**
 - Web/App/Social Media
 - Satellite
 - Location/Foot Traffic
- **另类数据与现实世界**
 - 挖掘时间地理数据对生态现象建模
- **另类数据与人类行为**
 - 手机数据与用户特征
 - 社交媒体与电子商务用户行为
- **另类数据与经济金融研究**
 - 挖掘卫星图片预测经济指标
 - 突发风险事件监测与市场建模

- **What:** 什么是另类数据？
- **Why:** 我们为什么要利用另类数据？
- **How:** 我们如何使用另类数据？

